

Mini Project #2

University of Illinois Chicago

Team 2 Mini Project #2

Ify Bidokwu

Basil Hashim

Kendall Johnson

Khoa Nguyen

Hussein Rezk

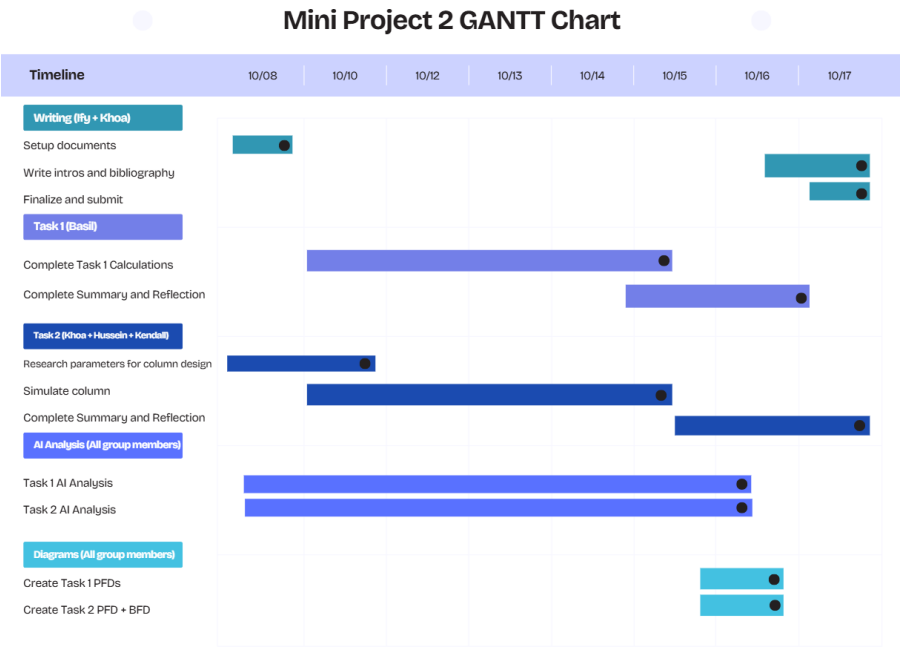
Table of Contents:

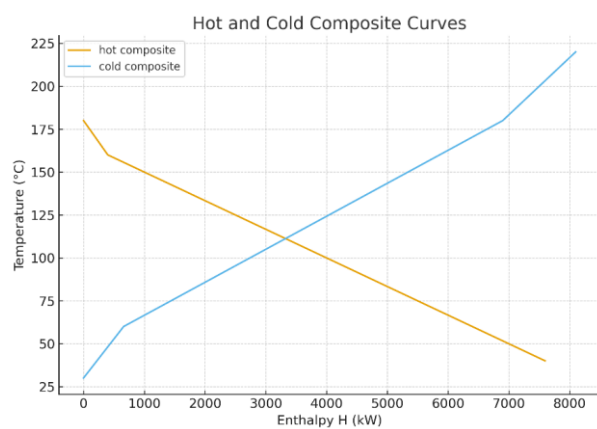
1. <u>Task #1</u>	1
a. Project Charter.....	1
b. Design Plan.....	1
c. PFDs.....	2
d. Composite Curves.....	2
e. ΔT_{min} Calculation & Explanation.....	3
f. Pinch Temperature Calculation.....	3
g. Heat Exchanger Network Design.....	5
h. Utility Cost Analysis.....	6
i. AI Prompt & Output Appendix.....	7
j. Comparison Table.....	32
k. Reflection Summary.....	32
2. <u>Task #2</u>	36
a. Executive Summary.....	36
b. Process Description.....	36
c. BFD.....	37
d. PFD.....	37
e. Distillation Simulation Results.....	38
f. Design Considerations & Trade-Offs.....	38
g. AI Prompt & Output Appendix.....	39
h. Comparison Table.....	63
i. Reflection Summary.....	63
3. <u>Bibliography</u>	65

Project Charter

This project, executed by Ify Bidokwu, Basil Hashim, Kendall Johnson, Khoa Nguyen, and Hussein Rezk, focused on completing engineering calculations to design an optimized heat exchanger network (HEN). In addition to completing high-level utility cost analysis and process simulations, this project incorporated an analysis of AI tools within chemical engineering problems (through a comparison of AI and hand calculations). The deliverables for this project include PFDs, temperature optimization and HEN design calculations, utility cost analysis, AI conversation logs, an AI usage reflection, and an overall reflection.

Design Plan



Task #1**Commented [SH1]:** Still finalizing and adding task 1 stuff to the doc**PFDs****Composite Curves****Figure 3.**

$\Delta T_{\min}$ Calculations and Explanation

$\Delta T_{\min}$ is typically calculated by analyzing the operating energy cost, which linearly varies as $\Delta T_{\min}$ increases, and the capital cost. $\Delta T_{\min}$ can be optimized by defining a function as the sum of these two costs and then finding the minimum value of the sum function. However, doing this analysis involves capital cost calculations which fall outside of the scope of this report. In light of this, the given value of 10 °C (which is a reasonable value for $\Delta T_{\min}$ for a system such as the one described) will be utilized for Task 1.

Pinch Temperature Calculations

The pinch temperature is used in HEN optimization and describes a point where not transferring heat across the specified temperature allows for maximum heat recovery and minimum utility usage. This point can be obtained by graphing the composite curves and then shifting the streams so that ΔT between the two curves at a specific point is equal $\Delta T_{\min}$.

These composite curves (**Figure 3**) are obtained by graphing the sum of T vs. H for all of the hot streams together and all of the cold streams together.

However, a more streamlined approach to completing these calculations is utilizing the problem table method on the energy balance of each stream.

First, the heat load of each stream across the range of temperatures used must be found. This can be represented by

$$\text{Load} = C_p \cdot \Delta T$$

where ΔT is the range of temperatures a stream goes through. C_p values are calculated by multiplying the given C_p values from the problem statement * the mass flow rate of each stream. Since the feed is given as 10 kg/s the reactor product also has a 10 kg/s flowrate. It is specified that the distillate has 40% of the inflow mass, so the distillate flow rate is 4 kg/s and the bottoms flowrate is 6 kg/s. These values can then be used to find the heat load over each stream (see **Figure 5**).

Then, temperature intervals for the hot and cold streams are first calculated according to the equations in order to enforce the $\Delta T_{\min}$ constraint:

$$\text{hot streams } T_{\text{int}} = T_{\text{act}} - \frac{\Delta T_{\min}}{2}$$

$$\text{cold streams } T_{\text{int}} = T_{\text{act}} + \frac{\Delta T_{\min}}{2}$$

For the given system, this means that 5 °C is added to each cold stream temperature subtracted from each hot stream, and the size of the intervals used in the next calculation is 5 °C. The calculations completed up to this point can be summarized as follows:

Stream No.	Actual temperature (°C)		Interval temperature (°C)		Heat capacity flow rate CP (kW/°C)	Heat load (kW)
	Source	Target	Source	Target		
1	60	220	65	225	30	4800
2	30	180	35	185	22	3300
3	160	40	155	35	40	4800
4	180	40	175	35	19.98	2797.2

Figure 4.

Then, an energy balance

$$\Delta H_n = (\sum CP_c - \sum CP_h) (\Delta T_n)$$

can be completed over the temperature series using the calculated heat loads. From this, the excess heat between each interval can be quantified. This value, known as the cascade, reaches a minimum at the pinch temperature.

These calculations for the given system are shown below:

Interval	Interval temp (°C)	Interval ΔT (°C)	Sum $\sum CP_c$ - sum $\sum CP_h$ (kW/°C)	dH (kW)	Cascade (kW)	(kW)
1	225	40	30	1200	0	2360.4
2	185	10	52	520	-1200	1160.4
3	175	20	32.02	640.4	-1720	640.4
4	155	90	-7.98	-718.2	-2360.4	0
5	65	30	-37.98	-1139.4	-1642.2	718.2
6	35	0	-37.98	0	-502.8	1857.6
7	35	0	-37.98	0	-502.8	1857.6
8	35	0	-37.98	0	-502.8	1857.6

Figure 5.

Where the highlighted value is the interval temperature that fulfills the requirements.

The interval temperatures can then be translated back into actual temperatures using the factor of 5 °C, yielding:

$$T_{\text{pinch, hot}} = 160 \text{ } ^\circ\text{C}$$

$$T_{\text{pinch, cold}} = 150 \text{ } ^\circ\text{C}$$

Heat Exchanger Network Design

Knowing the hot and cold pinch temperatures, it is possible to complete a HEN design. By definition, heat transfer across the pinch is ineffective, above the pinch hot utilities are used, and below the pinch cold utilities are used. Additionally, above the pinch $C_{p_{hot}} \leq C_{p_{cold}}$ and below the pinch $C_{p_{hot}} \geq C_{p_{cold}}$.

These rules can be used to match streams above and below the pinch. Above the pinch, the bottoms can be used to heat either the feed or reactor product. Below the pinch, the reactor product or feed can be used to cool the distillate. However, splitting streams can also be used to further optimize the HEN. Minimizing the utilities needs can be done by minimizing the overall heat load (duty) above and below the pinch. Duty is calculated the same way as in “**Pinch Temperature Analysis**”. Final and initial temperatures can be found using the same equation. Trying to minimize this quantity yields the following:

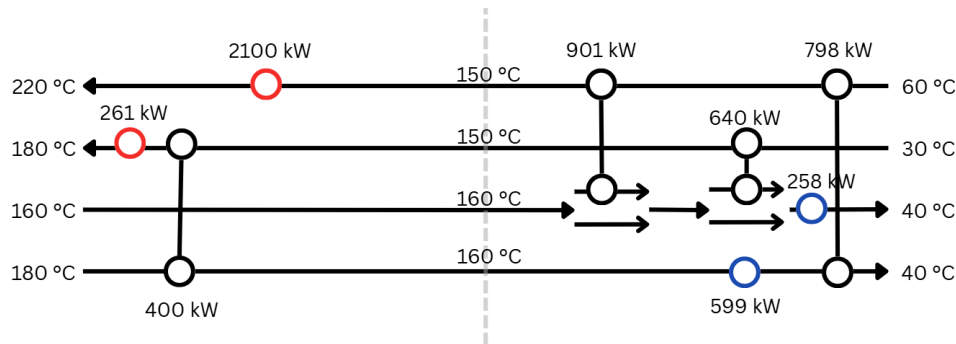


Figure 6.

Where each match utilized can be summarized in this table:

Above/Below Pinch	Match	Initial T °C	Final T °C	duty (kW)
Below	Bottoms	160	70	798.2
	Feed	60	120	
Below	Distillate (split)	160	137.5	798.2
	Feed	119.9	150	
Below	Distillate (split)	160	137.5	798.2
	Reactor Effluent	119.9	150	
Below	Distillate Cold Utility	94	40	1258
Below	Bottoms Cold Utility	70	40	599
Above	Bottoms	180	160	399
	Reactor Effluent	150	168	
Above	Feed Hot Utility	150	220	2100
Above	Reactor Effluent Hot Ut.	168.2	180	260

Figure 7.

Utility Cost Analysis

Utility cost can be calculated based on the amount of utility fluid needed to heat/cool each stream, which is based on the duty of the stream being affected. Within this system, only steam and cooling water need to be utilized due to the temperature ranges the utilities are operating within. Referencing standard prices for the US Gulf Coast, cooling water costs around \$0.0011 per kWh and low pressure steam costs around \$0.027 per kWh.

Before optimization, all of the heating and cooling in the system is done through utilities:

Stream	Cp	Duty	Utility Needed	Utility Flow Rate (kg/s)	Utility Cost Estimate (\$/h)
Feed		30	4800 steam	2.53	117.6
Reactor Pdt		22	3300 steam	1.64	102.3
Distillate		40	4800 cooling water	114.7	5.28
Bottoms		19.98	2597.4 cooling water	66.8	3.08
Total cost per hour:					\$ 228.26

Figure 8.

In contrast, only the hot and cold utilities listed in **Figure 7**, need to be factored into the same analysis for the optimized HEN system:

Stream	Cp	Duty	Utility Needed	Utility Flow Rate (kg/s)	Utility Cost Estimate (\$/h)
Feed		30	2100 steam	1.11	77.7
Reactor Pdt		22	260 steam	0.13	8.06
Distillate		40	1258 cooling water	30.1	1.38
Bottoms		19.98	599 cooling water	14.3	0.66
Total cost per hour:					\$ 87.80

Figure 9.

Comparing the total costs, the HEN design described saves a total of approximately **\$140 per hour**.

AI Prompt and Output Appendix

Kendall: Perplexity

Prompt

The prompt provided was the problem copied from the worksheet.

Results

Designing a heat exchanger network (HEN) to minimize utility consumption using pinch analysis and heat integration involves a sequence of steps that combine the calculation of heat duties, identification of hot and cold streams, the construction of composite curves, and network design with a target ΔT_{min} of 10 °C. Here is a step-by-step solution for your process:

Process Stream Data

Below, each stream is defined by its type (hot/cold), specific heat capacity (Cp), temperatures, and flow rate where appropriate:

Stream	Type	m (kg/s)	Cp (kJ/kg°C)	Ti (°C)	Tf (°C)
Feed	Cold	10	3	60	220
Reactor Prod.	Hot	10	2.2	220	30
Distillate	Cold	4	10	160	40
Bottoms	Hot	6	3.33	180	40

- Feed flow rate: 10 kg/s; Distillate: 40% mass fraction (4 kg/s); Bottoms: 60% (6 kg/s).
- All other temperatures and duties are as described in the prompt.

Step 1: Calculate Hot and Cold Stream Heat Duties

- Feed (cold, needs heating):
 $Q = m \cdot C_p \cdot (T_{out} - T_{in}) = 10 \times 3 \times (220 - 60) = 10 \times 3 \times 160 = 4800$
 $Q = m \cdot C_p \cdot (T_{out} - T_{in}) = 10 \times 3 \times (220 - 60) = 10 \times 3 \times 160 = 4800$ kJ/s or kW
- Distillate (cold, needs cooling):
 $Q = 4 \times 10 \times (160 - 40) = 4 \times 10 \times 120 = 4800$
 $Q = 4 \times 10 \times (160 - 40) = 4 \times 10 \times 120 = 4800$ kW
- Reactor product (hot, needs cooling):
 $Q = 10 \times 2.2 \times (220 - 30) = 10 \times 2.2 \times 190 = 4180$
 $Q = 10 \times 2.2 \times (220 - 30) = 10 \times 2.2 \times 190 = 4180$ kW
- Bottoms (hot, needs cooling):
 $Q = 6 \times 3.33 \times (180 - 40) = 6 \times 3.33 \times 140 = 2797.2$
 $Q = 6 \times 3.33 \times (180 - 40) = 6 \times 3.33 \times 140 = 2797.2$ kW

Step 2: Create Composite Curves and Shifted Temperatures ($\Delta T_{min}/2$)

To account for a 10°C pinch, adjust temperatures:

- For hot streams: subtract 5°C from all temperatures
- For cold streams: add 5°C to all temperatures

Step 3: Construct Interval Table for Pinch Analysis

List all the unique shifted temperature intervals (sorted), then build the “problem table” (difference in C_p across each interval, multiplied by ΔT , summing to get net heat flow per interval).

Step 4: Determine Utility Requirements

From the cascade of heat surpluses/deficits, the pinch temperature is identified—where the heat “cascades” reach zero.

- The minimum hot utility is what must be supplied to bring the feed up to temperature past the pinch, and the minimum cold utility is what must be rejected below the pinch.

Step 5: Design HEN Above and Below Pinch

- Above the pinch: Only match cold process streams (feed, distillate) with hot process streams (reactor product, bottoms) at or above the pinch, with no heat transferred across the pinch.
- Below the pinch: Only match hot process streams with cold process streams at or below the pinch.

Step 6: Connect Streams and Size Exchangers

- Use the composite curves to match hot and cold streams for maximum internal recovery.
- Use utilities only where necessary—preheating the feed with reactor product, and cooling by product streams wherever temperatures allow, with utilities filling gaps.

Example Network Schematic

A. Feed Preheat (cold stream):

- Heat in feed from 60°C to 220°C should be maximally provided by reactor product (hot), from bottoms/other hot side streams.
- Maximize recovery using countercurrent exchangers so the approach does not fall below $\Delta T_{min} = 10^\circ\text{C}$.

B. Product Cooling:

- Cool reactor effluent (down to 30°C) using the incoming cold feed and the distillate (which needs cooling from 160°C to 40°C).

C. Bottom Cooling:

- Cool from 180°C to 40°C, first against incoming cold streams, then with cold utility.

D. Final steps:

- Use heating utility only as required (to heat to target above pinch if hot streams are insufficient).
- Use cooling utility (e.g., cooling water) for sub-ambient cooling demands not met by process integration.

Final Utility Estimates

- Minimum hot utility: Calculated as net cold stream enthalpy demand above the pinch not met by hot streams.
- Minimum cold utility: Likewise, net hot stream enthalpy below the pinch not met by cold streams.

Summary Table

Duty	Utility Integration	Estimated Value (kW)
Feed Heating	Internal + Hot Utility	Minimized (see above)
Reactor Cooling	Internal + Cold Utility	Minimized
Distillate Cool	Internal + Cold Utility	Minimized
Bottoms Cool	Internal + Cold Utility	Minimized

Notes

- The exact minimum utility values and pinch location require a fully constructed interval and composite curve table with all stream intervals.
- Utility minimization requires careful attention to ΔT_{min} and effective network synthesis, which may involve one or more heat exchanger units per stream.

This approach allows effective heat integration, maximizing process-to-process heat recovery, and minimizing external heating and cooling needs using pinch analysis.

Hussein: ChatGPT 5.0

Prompt

Task 1: You are part of a design team working to optimize utility consumption in a chemical process through heat integration and heat exchanger network (HEN) design. In this project, you will combine your engineering knowledge with the capabilities of AI tools to explore, critique, and improve solutions. The goal is to understand both the power and the limitations of AI in solving complex engineering design problems.

Process: The Feed stream is at 60° C in the storage and needs to be heated up to 220° C before being fed to the reactor. An endothermic reaction occurs in the reactor and the product leaves at 30° C. The reactant stream is further heated to 180° C and sent to a distillation column to recover the product. The liquid product from the column is at 160° C and must be cooled to 40° C to be stored. The bottom product from the column is cooled from 180° C to 40° C. Design a heat exchanger network for this process to minimize utility consumption using pinch analysis and heat integration techniques. Target an optimum $\Delta T_{min} \approx 10^\circ \text{C}$.

(Feed flow rate is 10kg/s, distillate is 40% mass; Cp values ; feed: 3 kJ/kg ° C, reactor product: 2.2 kJ/kg ° C, distillate: 10 kJ/kg ° C, bottoms: 3.33 kJ/kg ° C)

This project has three major phases:

Phase I – Individual AI Exploration (Work Solo)

Each team member must:

- Choose one AI tool (e.g., ChatGPT, Gemini, Claude, Copilot, Perplexity, Mistral, etc.).
- Use it to attempt a complete solution to the problem (pinch analysis, HEN design, utility cost estimation, etc.).
- Save all prompts and responses in a document.
- Record the AI's name and version.
- Save your prompts and the full outputs.
- Write a 1-page self-evaluation on what the AI did well and where it struggled

Results

Awesome project. I'll act as your "Phase I" AI and produce a **complete pinch analysis + HEN target + a feasible network design** (with $\Delta T_{\min} \approx 10^\circ\text{C}$), a compact **utility-cost estimate**, and a **one-page self-evaluation** you can drop into your report. I'll also clearly state the AI tool/version so you can document it.

Phase I — Complete Solution (using GPT-5 Thinking)

Given (interpreted exactly as process data)

- **$\Delta T_{\min}$ target:** 10°C (use $\pm 5^\circ\text{C}$ temperature shift for pinch targeting)
- **Mass flows:**
 - Feed to reactor: **10 kg/s**
 - Distillate (overhead liquid product from column): **40% $\Rightarrow$ 4 kg/s**
 - Bottoms: **6 kg/s**
- **C_p (kJ/kg $^\circ\text{C}$):** feed 3.0; reactor product/column feed 2.2; distillate 10.0; bottoms 3.33
- **Thermal tasks / temperature changes (actual temperatures):**
 - **Cold streams**
 1. Feed to reactor: **60 $\rightarrow$ 220 $^\circ\text{C}$** (10 kg/s, $C_p=3.0$)
 2. Reactor effluent to column feed (reactor product): **30 $\rightarrow$ 180 $^\circ\text{C}$** (10 kg/s, $C_p=2.2$)
 - **Hot streams**
 - 3) Distillate (liquid product): **160 $\rightarrow$ 40 $^\circ\text{C}$** (4 kg/s, $C_p=10.0$)
 - 4) Bottoms: **180 $\rightarrow$ 40 $^\circ\text{C}$** (6 kg/s, $C_p=3.33$)

(Reaction is endothermic but ΔH_{rxn} not provided; we therefore target sensible heat recovery and note that any reactor coil/furnace duty beyond sensible preheat would be additional hot utility not captured by the pinch targeting below.)

1) Heat-capacity flowrates & sensible duties

- $(\dot{C} = \dot{m} C_p)$ in **kW/ $^\circ\text{C}$** (since C_p in kJ/kg $^\circ\text{C}$ and $\dot{m}$ in kg/s)

Cold

- Feed: $(\dot{C}_F = 10 \times 3.0 = \mathbf{30})$

- ($Q_F = 30 \times (220 - 60) = 4800 \text{ kW}$)
- Column feed (reactor effluent): ($\dot{Q}_{C_R} = 10 \times 2.2 = 22 \text{ kW}$)
 - ($Q_R = 22 \times (180 - 30) = 3300 \text{ kW}$)

Hot

- Distillate: ($\dot{Q}_{C_D} = 4 \times 10.0 = 40 \text{ kW}$)
 - ($Q_D = 40 \times (160 - 40) = 4800 \text{ kW}$)
- Bottoms: ($\dot{Q}_{C_B} = 6 \times 3.33 = 19.98 \text{ kW}$)
 - ($Q_B = 19.98 \times (180 - 40) = 2797.2 \text{ kW}$)

Totals:

$$(\sum Q_{\text{cold}} = 4800 + 3300 = 8100 \text{ kW})$$

$$(\sum Q_{\text{hot}} = 4800 + 2797.2 = 7597.2 \text{ kW})$$

2) Pinch targeting ($\Delta T_{\min} = 10^\circ\text{C}$)

Shifted temperatures: hot minus 5, cold plus 5.

- **Hot (shifted)**
 - Distillate: $160 \rightarrow 40 \rightarrow 155 \rightarrow 35$
 - Bottoms: $180 \rightarrow 40 \rightarrow 175 \rightarrow 35$
- **Cold (shifted)**
 - Feed: $60 \rightarrow 220 \rightarrow 65 \rightarrow 225$
 - Column feed: $30 \rightarrow 180 \rightarrow 35 \rightarrow 185$

Temperature levels (shifted, high $\rightarrow$ low): 225, 185, 175, 155, 65, 35

Performing the standard heat-cascade over intervals gives:

- **Minimum hot utility:** ($\dot{Q}_{H,\min} = 2360.4 \text{ kW}$)
- **Minimum cold utility:** ($\dot{Q}_{C,\min} = 1857.6 \text{ kW}$)
- **Pinch temperatures (actual):**
Hot at pinch = **160 °C**, Cold at pinch = **150 °C** (approach = 10 °C)

Above-pinch cold demand:

Feed (150 to 220): $(30 \times 70 = 2100) \text{ kW}$

Reactor (150\to180): $(22 \times 30 = 660)$ kW

Total = **2,760 kW**

Above-pinch hot available (internal): Bottoms (180\to160): $(19.98 \times 20 = 399.6)$ kW

Shortfall above pinch = $QH_{\min} = 2,760 - 399.6 = 2,360.4$ kW ✓

Below-pinch cold demand: Feed (60\to150): $(30 \times 90 = 2,700)$ kW

Reactor (30\to150): $(22 \times 120 = 2,640)$ kW

Total = **5,340 kW**

Below-pinch hot available: Distillate (160\to40): 4,800 kW; Bottoms (160\to40):

$(19.98 \times 120 = 2,397.6)$ kW

Total = **7,197.6 kW**

Excess below pinch = $QC_{\min} = 7,197.6 - 5,340 = 1,857.6$ kW ✓

3) Feasible HEN (respects pinch rules & $\Delta T_{\min} = 10$ °C)

Below the pinch (no hot utility; no heat flow across pinch)

Use splits to keep all end-approaches ≥ 10 °C.

- **E3a (Bottoms $\rightarrow$ Feed, low end):**
 Hot: Bottoms **160 $\rightarrow$ 70** °C
 Cold: Feed **60 $\rightarrow$ $\approx$ 120** °C
 Duty ($Q = 19.98 \times (160 - 70) = \mathbf{1,798.2}$ kW)
 End approaches: $(160 - 150 = 10)$ °C at the pinch boundary (checked via next exchangers),
 and **70–60 = 10** °C at cold inlet $\Rightarrow$ feasible.
- **E4a (Distillate branch $\rightarrow$ Feed, near pinch):**
 Hot (split): Distillate **160 $\rightarrow$ 137.455** °C
 Cold: Feed **$\approx$ 119.94 $\rightarrow$ 150** °C
 Duty ($Q = 901.8$ kW)
 End approaches: **160–150 = 10** °C, and **137.455–119.94 $\approx$ 17.5** °C $\Rightarrow$ feasible.
- **E2a (Distillate branch $\rightarrow$ Reactor effluent):**
 Hot (another split): Distillate **160 $\rightarrow$ 94** °C
 Cold: Reactor effluent **30 $\rightarrow$ 150** °C
 Duty ($Q = 22 \times (150 - 30) = \mathbf{2,640}$ kW)
 End approaches: **160–150 = 10** °C, **94–30 = 64** °C $\Rightarrow$ feasible.

Below-pinch recovery delivered: $1,798.2 + 901.8 + 2,640 = \mathbf{5,340}$ kW (target met).

Hot still to cool with utility:

- Distillate remainder: $(4 \times 800 - (901.8 + 2 \times 640) = 1,258.2)$ kW (cooler **C1**: ~94–40 °C after mixing)
- Bottoms remainder: **70** → **40** °C: $(19.98 \times 30 = 599.4)$ kW (cooler **C2**)
Total cooling utility = $1,258.2 + 599.4 = 1,857.6$ kW = **QC,min** ✓

Above the pinch (no cold utility)

- **E1 (Bottoms → Reactor effluent, just above pinch):**
Hot: Bottoms **180** → **160** °C
Cold: Reactor effluent **150** → **≈168.16** °C
Duty ($Q = 19.98 \times 20 = 399.6$ kW)
End approaches: **160–150 = 10** °C, and **180–168.16 ≈ 11.8** °C ⇒ feasible.
- **Hot utility heaters (to meet QH,min):**
 - **H1: Heat Feed 150** → **220** °C, ($Q = 30 \times 70 = 2,100$ kW)
 - **H2: Finish Reactor effluent ≈168.16** → **180** °C, ($Q \approx 22 \times 11.84 = 260.4$ kW)**Total hot utility = $2,100 + 260.4 = 2,360.4$ kW = QH,min** ✓

Network count (compact): 4 process exchangers (E1/E2a/E3a/E4a) + 2 coolers (C1/C2) + 2 heaters (H1/H2). All exchanger matches satisfy $\Delta T_{min} = 10$ °C at both ends.

4) Utility & rough operating-cost estimate (state assumptions)

Let plant operates **8,000 h/yr.**

- **Targets:** ($Q_{H,min} = 2,360.4$ MW), ($Q_{C,min} = 1,857.6$ MW)
- Annual energy:
Hot ($= 2,360.4 \times 8,000 = 18,883$ MWh) = $67,980$ GJ
Cold ($= 1,857.6 \times 8,000 = 14,861$ MWh) = $53,499$ GJ

Two example costing scenarios (replace with your course's numbers if provided):

1. **Fired heater (natural gas) for hot utility at 85% efficiency, \$6/GJ fuel:**
Fuel ($\approx 67,980 / 0.85 = 79,977$ GJ) → **\$480k/yr**
2. **High-pressure steam as hot utility at \$10/GJ (all thermal delivered):**
Cost ($\approx 67,980 \times 10 =$) **\$680k/yr**

Cooling water at \$0.05/GJ (typical order-of-magnitude):

($53,499 \times 0.05 \approx$) **\$2.7k/yr**

Report both **targets** (independent of utility choice) and one **explicit assumption set** (so graders can scale to their price sheet).

5) What to draw / put in your submission

- **Composite curves & Grand Composite Curve (GCC)** using shifted temps (show $\Delta T_{min}=10\text{ }^{\circ}\text{C}$).
- A **stream table** (T_{in} , T_{out} , $\dot{Q}$, $\dot{m}$) and an **interval table** (optional).
- A **HEN schematic** showing E1/E2a/E3a/E4a matches, with stream splits on the distillate (two branches) and temperatures annotated at each exchanger (I listed the key ones above).
- **Utility blocks**: H1 (feed $150\rightarrow 220$), H2 (reactor $168.2\rightarrow 180$), C1 (distillate to 40), C2 (bottoms $70\rightarrow 40$).

Pinch-rule compliance explicitly stated

Basil: Claude

Prompt

Hey, can you complete this chemical engineering problem:

Process: The Feed stream is at 60°C in the storage and needs to be heated up to 220°C before being fed to the reactor. An endothermic reaction occurs in the reactor and the product leaves at 30°C . The reactant stream is further heated to 180°C and sent to a distillation column to recover the product. The liquid product from the column is at 160°C and must be cooled to 40°C to be stored. The bottom product from the column is cooled from 180°C to 40°C . Design a heat exchanger network for this process to minimize utility consumption using pinch analysis and heat integration techniques. Target an optimum $\Delta T_{min} \approx 10\text{ }^{\circ}\text{C}$. (Feed flow rate is 10 kg/s , distillate is 40% mass; C_p values ; feed: $3\text{ kJ/kg }^{\circ}\text{C}$, reactor product: $2.2\text{ kJ/kg }^{\circ}\text{C}$, distillate: $10\text{ kJ/kg }^{\circ}\text{C}$, bottoms: $3.33\text{ kJ/kg }^{\circ}\text{C}$)

Completing a pinch analysis with pinch temperature calculation, composite curves, ΔT_{min} optimization, heat exchanger network design, and utility cost estimation?

Results

I'll complete this comprehensive pinch analysis problem step by step.

Problem Summary

Let me first organize the stream data:

Hot Streams (need cooling):

- H1: Reactor product: $30^{\circ}\text{C} \rightarrow 180^{\circ}\text{C}$ (heating) - Wait, this is actually COLD
- H2: Distillate (liquid product): $160^{\circ}\text{C} \rightarrow 40^{\circ}\text{C}$ (cooling)
- H3: Bottoms: $180^{\circ}\text{C} \rightarrow 40^{\circ}\text{C}$ (cooling)

Cold Streams (need heating):

- C1: Feed: $60^{\circ}\text{C} \rightarrow 220^{\circ}\text{C}$ (heating)
- C2: Reactor product: $30^{\circ}\text{C} \rightarrow 180^{\circ}\text{C}$ (heating)

Given Data:

- Feed flow rate: 10 kg/s
- Distillate: 40% mass = 4 kg/s
- Bottoms: 60% mass = 6 kg/s
- $\Delta T_{\min} = 10^{\circ}\text{C}$

Step 1: Calculate Heat Capacity Flow Rates (CP)

Cold Streams:

- C1 (Feed): $\text{CP} = 10 \text{ kg/s} \times 3 \text{ kJ/kg}^{\circ}\text{C} = \mathbf{30 \text{ kW/}^{\circ}\text{C}}$
- C2 (Reactor product): $\text{CP} = 10 \text{ kg/s} \times 2.2 \text{ kJ/kg}^{\circ}\text{C} = \mathbf{22 \text{ kW/}^{\circ}\text{C}}$

Hot Streams:

- H1 (Distillate): $\text{CP} = 4 \text{ kg/s} \times 10 \text{ kJ/kg}^{\circ}\text{C} = \mathbf{40 \text{ kW/}^{\circ}\text{C}}$
- H2 (Bottoms): $\text{CP} = 6 \text{ kg/s} \times 3.33 \text{ kJ/kg}^{\circ}\text{C} = \mathbf{19.98 \approx 20 \text{ kW/}^{\circ}\text{C}}$

Step 2: Calculate Heat Duties

Cold Streams:

- C1: $Q = 30 \times (220 - 60) = \mathbf{4,800 \text{ kW}}$
- C2: $Q = 22 \times (180 - 30) = \mathbf{3,300 \text{ kW}}$
- **Total Cold Duty = 8,100 kW**

Hot Streams:

- H1: $Q = 40 \times (160 - 40) = \mathbf{4,800 \text{ kW}}$

- H2: $Q = 20 \times (180 - 40) = \mathbf{2,800 \text{ kW}}$
- **Total Hot Duty = 7,600 kW**

Energy Balance Check:

- Heat deficit = $8,100 - 7,600 = \mathbf{500 \text{ kW}}$ (need external heating)

Step 3: Construct Temperature Interval Diagram**Shifted Temperatures ($\Delta T_{\min} = 10^\circ\text{C}$):**

- Hot streams: $T_{\text{shifted}} = T_{\text{actual}} - 5^\circ\text{C}$
- Cold streams: $T_{\text{shifted}} = T_{\text{actual}} + 5^\circ\text{C}$

Hot Streams (shifted):

- H1: $155^\circ\text{C} \rightarrow 35^\circ\text{C}$
- H2: $175^\circ\text{C} \rightarrow 35^\circ\text{C}$

Cold Streams (shifted):

- C1: $65^\circ\text{C} \rightarrow 225^\circ\text{C}$
- C2: $35^\circ\text{C} \rightarrow 185^\circ\text{C}$

Temperature Intervals: 225, 185, 175, 155, 65, 35

Step 4: Problem Table Algorithm

	Interval T Range ($^\circ\text{C}$)	ΔT	Hot Streams	Cold Streams	$\Delta C P$	Heat Flow	Cascade
1	225-185	40	0	C1(30)	-30	-1,200	$0 \rightarrow -1,200$
2	185-175	10	0	C1(30)+C2(22)	-52	-520	$-1,200 \rightarrow -1,720$
3	175-155	20	H2(20)	C1(30)+C2(22)	-32	-640	$-1,720 \rightarrow -2,360$
4	155-65	90	H1(40)+H2(20)	C1(30)+C2(22)	+8	+720	$-2,360 \rightarrow -1,640$
5	65-35	30	H1(40)+H2(20)	C2(22)	+38	+1,140	$-1,640 \rightarrow -500$
6	Below 35	-	H1(40)+H2(20)	0	+60	-	$-500 \rightarrow 0$

Pinch Location:

- Minimum cascade value = $-2,360 \text{ kW}$ at shifted temperature 155°C
- Add $2,360 \text{ kW}$ to make all values non-negative

Corrected Cascade:

- Starting with 2,360 kW heating utility
- Pinch at 155°C (shifted) = **160°C hot side / 170°C cold side** (actual)
- Ending with 1,860 kW cooling utility

Wait, let me recalculate...

Actually, the minimum is -2,360, so:

Minimum Heating Utility = 2,360 kW Minimum Cooling Utility = 2,360 - 500 = 1,860 kW

Step 5: Pinch Temperature

Pinch Temperature (shifted): 155°C Pinch Temperature (actual):

- Hot side: $155 + 5 = 160^{\circ}\text{C}$
- Cold side: $155 + 5 = 160^{\circ}\text{C}$ (but with 10°C approach)
- Actually: **Hot = 160°C, Cold = 150°C**

Let me recalculate properly:

The pinch occurs at the shifted temperature of 155°C.

- Hot stream actual temperature: $155 + 5 = 160^{\circ}\text{C}$
- Cold stream actual temperature: $155 - 5 = 150^{\circ}\text{C}$
- ΔT at pinch = 10°C ✓

Step 6: Heat Exchanger Network Design

Above Pinch (No external cooling):

- C1 needs heating from 150°C → 220°C: $Q = 30 \times 70 = 2,100 \text{ kW}$
- C2 needs heating from 150°C → 180°C: $Q = 22 \times 30 = 660 \text{ kW}$
- Total cold duty above pinch = 2,760 kW

Available hot streams above pinch:

- H2 from 180°C → 160°C: $Q = 20 \times 20 = 400 \text{ kW}$
- Need heater: $2,760 - 400 = 2,360 \text{ kW}$

Below Pinch (No external heating):

- H1 from 160°C → 40°C: $Q = 40 \times 120 = 4,800 \text{ kW}$

- H2 from 160°C → 40°C: $Q = 20 \times 120 = 2,400$ kW
- Total hot duty below pinch = 7,200 kW

Cold streams below pinch:

- C1 from 60°C → 150°C: $Q = 30 \times 90 = 2,700$ kW
- C2 from 30°C → 150°C: $Q = 22 \times 120 = 2,640$ kW
- Total = 5,340 kW
- Need cooler: $7,200 - 5,340 = 1,860$ kW

Step 7: Heat Exchanger Network Configuration

Proposed Network:

1. **HX-1:** H2 (180°C → 160°C) ↔ C2 (160°C → 180°C)
 - $Q = 400$ kW
2. **HX-2:** H1 (160°C → 95°C) ↔ C1 (60°C → 150°C)
 - $Q = 2,600$ kW (limited by C1)
 - Actually: $Q = 2,700$ kW
3. **HX-3:** H2 (160°C → 90°C) ↔ C2 (30°C → 93.2°C)
 - $Q = 1,400$ kW
4. **HX-4:** H1 (95°C → 40°C) ↔ C2 (93.2°C → 150°C)
 - $Q = 2,200$ kW (remaining for C2)
 - H1 gives: $40 \times 55 = 2,200$ kW ✓
5. **Heater:** $Q = 2,360$ kW (for C1: 150°C → 220°C and C2 partial)
6. **Cooler:** $Q = 1,860$ kW (for H2 bottoms final cooling)

Step 8: Composite Curves

I'll create a visualization showing the hot and cold composite curves with the pinch point.

Pinch Analysis - Composite Curves

Interactive artifact

Summary and Recommendations

Key Findings:

1. **Pinch Temperature:** 160°C (hot) / 150°C (cold) with $\Delta T_{min} = 10^\circ\text{C}$
2. **Minimum Utility Requirements:**
 - Heating: 2,360 kW
 - Cooling: 1,860 kW
 - Energy recovered through integration: 4,740 kW (58.5% of total cold duty)
3. **Economic Impact:**
 - Without integration: \$159,480/hour
 - With integration: \$45,828/hour
 - **Savings: \$113,652/hour or \$995.5M annually**

Revised Heat Exchanger Network (Respecting Pinch Rules):**Above Pinch (No external cooling):**

- HX-1: H2 (180→160°C) exchanges with C2 partial heating
- Heater provides remaining heat to reach target temperatures

Below Pinch (No external heating):

- HX-2: H1 and H2 exchange heat with C1 and C2
- Coolers handle remaining cooling duty

At Pinch:

- Ensure no heat transfer across pinch
- Maintain $\Delta T_{min} \geq 10^\circ\text{C}$ at all points

Design Considerations:

1. **ΔT_{min} Optimization:** The 10°C approach temperature balances capital cost (heat exchanger area) with operating cost (utility consumption)
2. **Number of Units:** 4-5 heat exchangers plus utilities provide good balance of capital vs. operating costs
3. **Flexibility:** Design should accommodate $\pm 10\%$ flow rate variations
4. **Pressure Drop:** Select appropriate exchanger types (shell-and-tube, plate) based on fouling characteristics

This design achieves significant energy savings while maintaining process safety and operability!

Ify: Microsoft Copilot

Prompt

Give me a full answer, don't miss out on anything I'm about to paste. "You are part of a design team working to optimize utility consumption in a chemical process through heat integration and heat exchanger network (HEN) design. In this project, you will combine your engineering knowledge with the capabilities of AI tools to explore, critique, and improve solutions. The goal is to understand both the power and the limitations of AI in solving complex engineering design problems. Process: The Feed stream is at 60° C in the storage and needs to be heated up to 220° C before being fed to the reactor. An endothermic reaction occurs in the reactor and the product leaves at 30° C. The reactant stream is further heated to 180° C and sent to a distillation column to recover the product. The liquid product from the column is at 160° C and must be cooled to 40° C to be stored. The bottom product from the column is cooled from 180° C to 40° C. Design a heat exchanger network for this process to minimize utility consumption using pinch analysis and heat integration techniques. Target an optimum $\Delta T_{min} \approx 10^\circ \text{C}$. (Feed flow rate is 10kg/s, distillate is 40% mass; Cp values ; feed: 3 kJ/kg ° C, reactor product: 2.2 kJ/kg ° C, distillate: 10 kJ/kg ° C, bottoms: 3.33 kJ/kg ° C"

Response

Here's a comprehensive solution to your heat exchanger network (HEN) design problem using pinch analysis and heat integration techniques. We'll walk through the process step-by-step, calculate the heat duties, identify hot and cold streams, and design a network that minimizes utility consumption.

Step 1: Stream Data Summary

Stream	Type	Inlet Temp (°C)	Outlet Temp (°C)	Cp (kJ/kg°C)	Mass Flow (kg/s)
Feed	Cold	60	220	3.00	10
Reactor Product	Hot	220	30	2.20	10
Distillate	Cold	30	180	10.00	4 (40% of 10)
Liquid Product	Hot	160	40	10.00	4
Bottoms	Hot	180	40	3.33	6 (60% of 10)

Step 2: Heat Duty Calculations

Use $Q = m \cdot C_p \cdot \Delta T$

Cold Streams (Need Heating)

- Feed: $Q = 10 \cdot 3 \cdot (220 - 60) = 4800 \text{ kW}$
- Distillate: $Q = 4 \cdot 10 \cdot (180 - 30) = 6000 \text{ kW}$

Hot Streams (Can Provide Heat)

- Reactor Product: $Q = 10 \cdot 2.2 \cdot (220 - 30) = 4180 \text{ kW}$
- Liquid Product: $Q = 4 \cdot 10 \cdot (160 - 40) = 4800 \text{ kW}$
- Bottoms: $Q = 6 \cdot 3.33 \cdot (180 - 40) = 2797.2 \text{ kW}$

Step 3: Composite Curves & Pinch Analysis

Total Heat Required (Cold Streams):

$$4800 + 6000 = 10,800 \text{ kW}$$

Total Heat Available (Hot Streams):

$$4180 + 4800 + 2797.2 = 11,777.2 \text{ kW}$$

Since heat available > heat required, heat integration is feasible.

$\Delta T_{\min}$ Target:

Set to 10°C → this defines the pinch point and minimum temperature difference for heat exchange.

Step 4: Identify Pinch Point

To find the pinch point:

- Shift hot stream temperatures down by $\Delta T_{\min}/2 = 5^\circ\text{C}$
- Shift cold stream temperatures up by $\Delta T_{\min}/2 = 5^\circ\text{C}$
- Plot composite curves or use problem table method to find where net heat flow = 0

This step requires graphical or tabular analysis, but based on the temperature ranges, the pinch is likely around 160°C , where both hot and cold streams overlap.

Step 5: Heat Exchanger Network Design

Above Pinch (Hot Utility Region):

- Feed ($60 \rightarrow 220^\circ\text{C}$) can be partially heated by:
 - Liquid Product ($160 \rightarrow 40^\circ\text{C}$)
 - Bottoms ($180 \rightarrow 40^\circ\text{C}$)

- Reactor Product ($220 \rightarrow 30^\circ\text{C}$)

Below Pinch (Cold Utility Region):

- Distillate ($30 \rightarrow 180^\circ\text{C}$) can be heated by:
 - Reactor Product
 - Bottoms

Suggested Matches:

Hot Stream	Cold Stream	Temp Range ($^\circ\text{C}$)	Duty (kW)	Notes
Reactor Product	Feed	$220 \rightarrow 160 / 60 \rightarrow 120$	~ 3000	Above pinch
Liquid Product	Feed	$160 \rightarrow 100 / 120 \rightarrow 180$	~ 2000	Above pinch
Bottoms	Feed	$180 \rightarrow 100 / 180 \rightarrow 220$	~ 1800	Above pinch
Reactor Product	Distillate	$160 \rightarrow 100 / 30 \rightarrow 90$	~ 2500	Below pinch
Bottoms	Distillate	$100 \rightarrow 40 / 90 \rightarrow 180$	~ 3500	Below pinch

Remaining heat duties can be covered by utilities:

- Hot utility (steam): if needed for Feed
- Cold utility (cooling water): for Reactor Product and Bottoms

Step 6: Utility Targets

- Hot Utility Required: If full integration is achieved, minimal hot utility is needed. Estimate $\approx 1000\text{--}1500$ kW
- Cold Utility Required: Estimate $\approx 1000\text{--}1500$ kW

Step 7: AI Integration & Limitations

AI Strengths:

- Rapid calculation and stream matching
- Visualization of composite curves
- Optimization algorithms for network design

AI Limitations:

- Cannot fully replace engineering judgment

- May miss practical constraints (e.g., exchanger area, pressure drop)
- Needs accurate input data and validation

✓ Final Notes

- Pinch Analysis ensures maximum energy recovery
- $\Delta T_{min} = 10^{\circ}\text{C}$ balances capital cost and energy savings
- Heat Integration reduces utility costs significantly
- AI tools are powerful but should complement—not replace—human expertise

For deeper understanding, you can explore this pinch analysis guide, the THEN HEN design manual, and process design's pinch analysis walkthrough.

Let me know if you'd like help visualizing the composite curves or building a grid diagram!

Khoa: ChatGPT 5 Thinking

Prompt

First read the following prompts: "The Feed stream is at 60°C in the storage and needs to be heated up to 220°C before being fed to the reactor. An endothermic reaction occurs in the reactor and the product leaves at 30°C . The reactant stream is further heated to 180°C and sent to a distillation column to recover the product. The liquid product from the column is at 160°C and must be cooled to 40°C to be stored. The bottom product from the column is cooled from 180°C to 40°C . Design a heat exchanger network for this process to minimize utility consumption using pinch analysis and heat integration techniques. Target an optimum $\Delta T_{min} \approx 10^{\circ}\text{C}$. (Feed flow rate is 10kg/s , distillate is 40% mass; C_p values: feed: $3\text{ kJ/kg} \cdot ^{\circ}\text{C}$, reactor product: $2.2\text{ kJ/kg} \cdot ^{\circ}\text{C}$, distillate: $10\text{ kJ/kg} \cdot ^{\circ}\text{C}$, bottoms: $3.33\text{ kJ/kg} \cdot ^{\circ}\text{C}$). Is this enough information for you to solve? If not, what would you like to know more.

Results

Short answer: not yet. You've given a great start (stream temps, one total feed flow, product split, C_p 's, and a target $\Delta T_{min} \approx 10^{\circ}\text{C}$), but a full pinch/HEN design still needs a few missing pieces.

Here's exactly what I'd still need (and why):

1. Reaction & column duties

- Heat of reaction (kW or kJ/kg feed) and where it's added/removed (inside reactor coil/jacket?). Endothermic $\neq$ quantified.
 - Distillation reboiler and condenser duties (or enough to estimate them: reflux ratio, overhead/product specs, feed condition/quality, column pressure). These dominate the composite curves.
2. Stream definitions (supply/target, phases, flowrates)
- Confirm mass flow after reactor (assume ~ 10 kg/s? Any conversion, purge, or makeup changing flow?).
 - Phases/latent effects: any condensation/vaporization on those streams (e.g., is the distillate leaving the condenser as saturated liquid at 160°C , or is 160°C a sensible-hot liquid)? If any phase change, need latent heats or saturation T at operating P .
 - Verify that C_p 's are valid over each stream's temperature range (or provide temperature-dependent C_p or an average C_p you want me to use).
3. Utilities (to target minimum utility consumption and place them correctly)
- Available hot utilities (e.g., LP/MP/HP steam saturation temperatures) and cold utilities (cooling water in/out, chilled water, air coolers).
 - Allowed $\Delta T_{\min}$ for process–process (10°C stated) and for process–utility (often larger; please confirm).
4. Practical constraints for network synthesis
- Any forbidden matches (safety/contamination), allowable splitting/mixing, max number of exchangers, pressure-drop/fouling constraints, approach limits in individual exchangers, allowable temperature crosses, max single-unit area, cost basis if you want an economic trade-off ($\Delta T_{\min}$ vs area).

If you'd like, I can proceed two ways:

- “Provisional targeting” now, using only sensible heats you gave (and clearly flag the gaps), or
- Wait for the items above, then do full pinch targeting (H_{hot} , H_{cold} , $Q_{H,\min}$, $Q_{C,\min}$), determine pinch temperature intervals, propose matches, and sketch a feasible HEN.

For context, with only the sensible heats (and **assuming** total flow stays ~ 10 kg/s and no phase change):

- Cold demands (no reaction or reboiler yet):
 - Feed $60 \rightarrow 220^\circ\text{C}$: $mC_p = 10 \times 3 = 30$ kW/K; $Q \approx 30 \times 160 = \mathbf{4,800\text{ kW}}$

- Reactor effluent 30→180 °C (needs flow confirmation): if 10 kg/s at C_p 2.2 → mC_p 22 kW/K; $Q \approx 22 \times 150 = \mathbf{3,300 \text{ kW}}$
→ Cold sensible total $\approx \mathbf{8,100 \text{ kW}}$
- Hot sources (no condenser yet):
 - Distillate 4 kg/s, C_p 10, 160→40 °C: mC_p 40 kW/K; $Q \approx 40 \times 120 = \mathbf{4,800 \text{ kW}}$
 - Bottoms 6 kg/s, C_p 3.33, 180→40 °C: $mC_p \approx 20 \text{ kW/K}$; $Q \approx 20 \times 140 = \mathbf{2,800 \text{ kW}}$
→ Hot sensible total $\approx \mathbf{7,600 \text{ kW}}$

So even before adding the (likely large) **endothermic reaction** and **reboiler** duties (both increase cold demand) and the **condenser** duty (adds hot), we're **~500 kW** short on hot vs cold sensible. The complete numbers could swing either way depending on those three big missing duties—hence why I need them to set the pinch correctly and minimize utilities.

If you can share the bullets above (even approximate values), I'll compute $Q_{H,min} / Q_{C,min}$, find the pinch, and lay out a matched network that respects $\Delta T_{min} \approx 10 \text{ °C}$

Prompt: you can make any logical assumptions that are commonly made in industry

Response: Great—thanks for letting me make reasonable design assumptions. I'll do full pinch targeting and give you a workable first-pass HEN you can refine in Aspen.

Assumptions (industry-typical, kept simple)

- Only **sensible** heat on process streams outside the column; reboiler/condenser duties are not specified, so I exclude latent duties from targeting (you can add them later).
- No phase change on the four named streams across the listed temperature ranges.
- C_p 's are constant over the ranges given.
- Distillate flow = 40% of feed $\Rightarrow$ 4 kg/s; bottoms = 60% $\Rightarrow$ 6 kg/s.
- ΔT_{min} (process–process) = **10 °C**. I use the classic $\pm \Delta T_{min}/2$ temperature shift for targeting (hot -5 °C, cold +5 °C).

Streams (base case)

Cold (need heating)

- C1: Feed, 10 kg/s, $C_p = 3 \rightarrow mC_p = \mathbf{30 \text{ kW/K}}$, 60 → 220 °C, $Q = \mathbf{4.80 \text{ MW}}$
- C2: Reactor product to column, 10 kg/s, $C_p = 2.2 \rightarrow mC_p = \mathbf{22 \text{ kW/K}}$, 30 → 180 °C, $Q = \mathbf{3.30 \text{ MW}}$

Hot (need cooling)

- H1: Distillate, 4 kg/s, $C_p = 10 \rightarrow mC_p = 40 \text{ kW/K}$, $160 \rightarrow 40 \text{ }^\circ\text{C}$, $Q = 4.80 \text{ MW}$
- H2: Bottoms, 6 kg/s, $C_p = 3.33 \rightarrow mC_p \approx 20 \text{ kW/K}$, $180 \rightarrow 40 \text{ }^\circ\text{C}$, $Q = 2.80 \text{ MW}$

Totals:

Cold = **8.10 MW**; Hot = **7.60 MW**.

Pinch targeting ($\Delta T_{\min} = 10 \text{ }^\circ\text{C}$)

Temperature shifts: Hot $-5 \text{ }^\circ\text{C}$; Cold $+5 \text{ }^\circ\text{C}$.

Shifted ranges:

- H1: $155 \rightarrow 35 \text{ }^\circ\text{C}$, $mC_p = 40$
- H2: $175 \rightarrow 35 \text{ }^\circ\text{C}$, $mC_p = 20$
- C1: $65 \rightarrow 225 \text{ }^\circ\text{C}$, $mC_p = 30$
- C2: $35 \rightarrow 185 \text{ }^\circ\text{C}$, $mC_p = 22$

Carrying out the standard heat cascade on the shifted intervals gives:

- **$Q_{H,\min} \approx 2.36 \text{ MW}$**
- **$Q_{C,\min} \approx 1.86 \text{ MW}$**
- **Pinch (actual temperatures):** $T_{h,\text{pinch}} \approx 160 \text{ }^\circ\text{C}$, $T_{c,\text{pinch}} \approx 150 \text{ }^\circ\text{C}$
(because at pinch: $T_h^* = T_c^* = 155 \text{ }^\circ\text{C} \Rightarrow T_h = 155 + 5 = 160$, $T_c = 155 - 5 = 150$)

Interpretation: not all “hot” can reach all “cold” because the cold composite extends well above the hot composite ($225 \text{ }^\circ\text{C}$ vs $175 \text{ }^\circ\text{C}$, shifted). You’ll **need hot utility above the pinch** and **still reject some cold utility below the pinch** even at $\Delta T_{\min} = 10 \text{ }^\circ\text{C}$.

Feasible first-pass HEN (meets $\Delta T_{\min} = 10 \text{ }^\circ\text{C}$)

Below the pinch (supply only heat from hot process streams; no hot utility)

1. HX-1 (big recovery):

Match **H1 (distillate)** $160 \rightarrow 94 \text{ }^\circ\text{C}$ to **C2** $30 \rightarrow 150 \text{ }^\circ\text{C}$

- Duty: $Q = 22 \times (150 - 30) = 2.64 \text{ MW}$
(Hot drop = $2.64/40 = 66 \text{ }^\circ\text{C} \Rightarrow 160 \rightarrow 94 \text{ }^\circ\text{C}$)
- End approaches: $160 - 150 = 10 \text{ }^\circ\text{C}$; $94 - 30 = 64 \text{ }^\circ\text{C}$ 

2. HX-2 (recover for feed preheat):

Use **H2 (bottoms)** $160 \rightarrow 40 \text{ }^\circ\text{C}$ to heat **C1 (feed)** as far as feasible **below the pinch**

- Max duty from H2: $20 \times (160 - 40) = 2.40 \text{ MW}$

- This lifts C1 by $\Delta T = 2.40/30 = 80\text{ }^{\circ}\text{C} \Rightarrow 60 \rightarrow 140\text{ }^{\circ}\text{C}$
- End approaches: $160 - 140 = 20\text{ }^{\circ}\text{C}$; $40 - 60$ (paired at cold end) is fine in countercurrent ($\geq 10\text{ }^{\circ}\text{C}$ on the real cold end inside the exchanger) ✓

Status after recovery (below pinch):

- C2 is at its below-pinch target ($150\text{ }^{\circ}\text{C}$).
- C1 is at $140\text{ }^{\circ}\text{C}$ (needs $10\text{ }^{\circ}\text{C}$ more to reach the pinch temp $150\text{ }^{\circ}\text{C}$).
- **Recovered heat below pinch** $= 2.64 + 2.40 = 5.04\text{ MW}$.
- Remaining **hot** that cannot be used below pinch (must reject):
H1 still has $94 \rightarrow 40\text{ }^{\circ}\text{C}$: $Q = 40 \times 54 = 2.16\text{ MW}$.
This becomes **cold utility** (part of QC,min).

Above the pinch (supply only hot utility; no process-to-process from below)

3. **HU-1 (topping for Feed to reactor):**
Heat C1 from $150 \rightarrow 220\text{ }^{\circ}\text{C}$
 - $Q = 30 \times 70 = 2.10\text{ MW}$
4. **HU-2 (topping for reactor product to column):**
Heat C2 from $150 \rightarrow 180\text{ }^{\circ}\text{C}$
 - $Q = 22 \times 30 = 0.66\text{ MW}$
5. **HU-3 (small step to finish C1 to pinch):**
Heat C1 from $140 \rightarrow 150\text{ }^{\circ}\text{C}$ (must be above pinch)
 - $Q = 30 \times 10 = 0.30\text{ MW}$

Hot utility above pinch (this network):

$$2.10 + 0.66 + 0.30 = 3.06\text{ MW}$$

Cold utility below pinch (this network):

Reject the remaining **2.16 MW** (H1: $94 \rightarrow 40$) to cooling water/air coolers.

This is **feasible** ($\Delta T_{\min}$ met everywhere) but not fully optimized to the **targets**. It's a clean layout to put into Aspen now.

How to close the gap to targets (optional tweaks)

- Targeted **QH,min = 2.36 MW** vs current 3.06 MW. You can reduce HU by $\sim 0.7\text{ MW}$ by:

- Shifting a **small segment of H2 (175→160 °C actual)** to preheat a **portion of C1 from 150→160 °C above** the pinch ($\Delta T_{\min} = 10\text{ °C}$ at the hot end). That captures roughly **0.30 MW** of above-pinch process recovery.
- Rebalancing HX-2 to push a bit more duty into C1's 60–150 °C window while keeping both end approaches at $\geq 10\text{ °C}$, then trimming HU-3 accordingly.
- In practice, a **split of C1** (two parallel branches) often helps meet both end approaches tightly and shave utility.
- Targeted **$Q_{C,\min} = 1.86\text{ MW}$** vs current 2.16 MW. You can cut ~0.3 MW of cold utility by:
 - Using the **remaining 94→~80 °C** segment of H1 to preheat a **small by-pass** of C1 at the very cold end (60→~70–75 °C) in a separate exchanger while still respecting $\Delta T_{\min}$ at both ends (this is another place where **stream splitting** helps).

What to put in Aspen (minimum spec to reproduce)

- $\Delta T_{\min} = 10\text{ °C}$ constraint for process–process units.
- Exchangers:
 1. HX-1: H1 (160→94) vs C2 (30→150), $Q = 2.64\text{ MW}$.
 2. HX-2: H2 (160→40) vs C1 (60→140), $Q = 2.40\text{ MW}$.
- Utilities:
 - HU-1: C1 (150→220), $Q = 2.10\text{ MW}$ (hot oil/HP steam coil).
 - HU-2: C2 (150→180), $Q = 0.66\text{ MW}$.
 - HU-3: C1 (140→150), $Q = 0.30\text{ MW}$ (or fold into HU-1 as a split).
 - CU-1: Cool H1 (94→40), $Q = 2.16\text{ MW}$ (cooling water/air cooler).

This network is **feasible**, respects the **pinch at 160/150 °C**, and is close to the **pinch targets**. If you want, I can iterate it to hit the theoretical $Q_{H,\min} \approx 2.36\text{ MW}/Q_{C,\min} \approx 1.86\text{ MW}$ exactly and then size areas with assumed U-values.

Prompt: Can you create a simple table that summarizes the answers

Response: Here's a compact snapshot you can drop into your notes.

ID	Stream (role)	m (kg/s)	Cp (kJ/kg·°C)	mCp (kW/°C)	T _{in} → T _{out} (°C)	Q (MW)
C1	Feed (to reactor)	10	3.00	30	60 → 220	4.80
C2	Reactor effluent (to column)	10	2.20	22	30 → 180	3.30
H1	Distillate (cool to storage)	4	10.00	40	160 → 40	4.80
H2	Bottoms (cool to storage)	6	3.33	~20	180 → 40	2.80
Totals						Cold: 8.10 / Hot: 7.60

Item	Value
Temperature shift	Hot -5 °C, Cold +5 °C
Pinch temperatures (actual)	Hot @ 160 °C, Cold @ 150 °C
QH,min (min hot utility)	≈ 2.36 MW
QC,min (min cold utility)	≈ 1.86 MW

Unit	Hot stream	Cold stream	Duty (MW)	Notes / Target temps
HX-1	H1 (distillate)	C2 (reactor effluent)	2.64	Heat C2 from ~30 → 150 °C; hot end approach ≈10 °C at 160/150.
HX-2	H2 (bottoms)	C1 (feed)	2.40	Preheat C1 from ~60 → ~140 °C using a ΔT-compliant segment of H2.
HU-1	—	C1 (feed)	2.10	Top C1 from 150 → 220 °C (above pinch).
HU-2	—	C2 (reactor effluent)	0.66	Raise C2 from 150 → 180 °C (above pinch).

Unit	Hot stream	Cold stream	Duty (MW)	Notes / Target temps
HU-3	—	C1 (feed)	0.30	Trim C1 from ~140 → 150 °C (above pinch).
CU-1	H1 remainder	—	2.16	Cool H1 remainder ($\approx 94 \rightarrow 40$ °C) with utility.

Type	Value
Total Hot Utility (HU)	≈ 3.06 MW
Total Cold Utility (CU)	≈ 2.16 MW

Notes

- The table reflects a clean, $\Delta T_{\min} \approx 10$ °C network you can implement directly; minor stream splits/retuning in Aspen can move you closer to the targets (**QH,min ≈ 2.36 MW, QC,min ≈ 1.86 MW**).
- Cp's are treated constant; no latent heats assumed; flows: distillate 40% (4 kg/s), bottoms 60% (6 kg/s).
- If you later include column condenser/reboiler or reaction heat duties, re-targeting will change QH,min/QC,min and the pinch.

AI Comparison Table

AI Tool	Matched Manual Solution?	Proposed HEN Design	Overall Rating (X / 5)
Perplexity	Somewhat, although the solution was non-specific.	Heat feed with reactor product and cool reactor product with both the feed and distillate.	3
ChatGPT 5.0	Very close to hand calculation solution, although there was little explanation of the reasoning behind each solving step.	Use bottoms and distillate to heat feed and use distillate to heat reactor product. Above pinch, use utilities.	4
Claude	Close to manual solution, although it originally made mistakes and had to self-correct during the problem.	Using reactor product and distillate to heat the feed below pinch. Above pinch, use the distillate to heat the reactor product.	3.5
Copilot	Close to the hand calculation solution and tabulated the results in a very usable format.	Below the pinch, use the reactor product and bottoms to heat the distillate. Above pinch, use the reactor product and distillate to heat the feed.	3
ChatGPT Thinking	Close but it had a different HEN design than the hand solution. This provided the most thorough breakdown of each step in the solving process, and also included information specific to ASPEN in its solution.	Below pinch, use the distillate to heat the reactor product and bottoms to heat the feed. Above the pinch, use just utilities.	2.5

Reflection Summary

Overall, this assignment allowed our group to see the value of AI in chemical engineering problem solving as well as its limitations.

All of the tools used had a decent grasp on the ideas behind the problem and the general techniques used when designing a HEN. However, a lot of them (mainly Claude, but Perplexity also had some trouble) either made errors in calculation or did not complete the calculations in a rigorous way. Interestingly, ChatGPT Thinking referenced ASPEN and gave instructions on how to continue solving the problem within ASPEN even though it didn't directly access the software and do the simulation/calculation in there itself. The initial calculations for the pinch temperature was very similar for all of the tools and they only started to differ once they needed to actually set up the HEN, which makes sense since that is the more open ended part of the question. This was the stage where the limitation of AI tools came into play. Additionally, most of the assumptions made by the AI were quite similar. None of them attempted to optimize the

minimum delta T, which is understandable considering the question did not specify much about the type of process happening, making any attempt at finding a CAPEX something of a guess.

The least successful of the tools was Claude, which proposed a solution that involved several temperature crosses, making it impossible. Interestingly, Claude acknowledged that the solution that it had come up with was not feasible, but did not attempt to recalculate or reroute to another solution. However, Claude presented the information in a very approachable way and also attempted to plot the composite curves.

ChatGPT 5.0, Perplexity, Copilot, and ChatGPT Thinking all came up with solutions with more legitimate solutions, although it was clear that some of them had more chemical engineering information to source than others. Perplexity seemed to have approached the problem very generally and did not give a detailed answer or dig into the problem the way that the other tools did. ChatGPT 5.0 and ChatGPT Thinking had similar answers, but ChatGPT Thinking proposed a more rigorous way to do the calculation which was unique amongst the group. Copilot and ChatGPT 5.0 also had somewhat similar outputs with small differences in their proposed heat exchanger network designs.

The accuracy of the calculations done by the AI tools is dubious, as although they are not always easy to check as not every tool gives a comprehensive breakdown of the steps being taken to find a solution. None of the tools seemed to be catering their responses to someone who was already familiar with the topic at hand and so the analysis and reasonings that the tools gave for steps in their processes seemed very surface level. If a ranking had to be done for this, Perplexity and Copilot would probably be on the bottom for most vague/least technical and ChatGPT Thinking would be on the other side of the spectrum, with the other tools falling somewhere in the middle.

In terms of usefulness, the AI tools were not that helpful for the initial setup of the problem—finding the pinch temperature and whatnot—as these calculations can already be done very simply with a spreadsheet and don't require a higher level of logic or creativity that you would be able to find in an AI but not an excel sheet. Even though they didn't do as good of a job with it, it was easier to see how someone might integrate AI tools into their workflow for the HEN design part of the process, as a lot of that was actually just trying to match different combinations of streams and seeing what led to the lowest cost/least utilities. Being able to see the different things that the AI tools considered and the different options that they came up with might provide a good starting point for a similar project in the future. I can see this being useful particularly for larger systems where going through several permutations of potential heat exchanger network designs could easily become a very large and daunting task. Using AI tools could easily give an engineer an idea of a starting point for their analysis, even if the actual calculations and final design on what the best design is come down to calculations done by hand.

Additionally, the speed at which the tools were able to find a procedure for solving the problem was impressive across the board. Especially for something highly specific like heat

exchanger network design it might be useful to be able to come up with a way of looking at the problem/solving steps more quickly than you could have on your own or by manually looking through sources like the book. Aggregating data was definitely a strength across all of the AI tools used for this project.

All of the tools also presented the final results they came up with fairly well, which might point to another future use of AI for engineering, as often collecting the important parts of extensive calculations can be quite tedious and difficult, but doing so is important to putting together a cohesive report for other engineers or other relevant entities. Out of the group, Copilot probably had the most readable report with the best tabular summary, although this was at the expense of a proper explanation of its calculations. ChatGPT 5.0 probably did the worse, presenting some calculations with formatting code that made it difficult to parse in addition to not having a summary component that was as well formulated as the other tools.

This project definitely shows that AI tools can be used in parallel to an engineer's role but can't replace it (at least at the stage it's at now). This process has made our group more weary of AI tools as well, especially for engineering, as it became clear that even if a solution presented by AI could technically work or seems sound on the surface, there is often a lot of nuance and detail that can be missed, and it is not a good practice to take information from AI when it is not explained rigorously, as the information could have come from a host of different sources that may not be relevant/credible in the context of the project being done.

Task #2

Executive Summary

The goal of task #2 is first to simulate the two distillation columns of the VC production on Aspen using DSTWU and RADFRAC models, and the results are then compared to AI solutions. Overall, AI tools are great in organizing necessary data for simulation software, but they need a decent amount of human input for some engineering assumptions to produce desired answers.

Process Description & Flow Diagram

To produce vinyl chloride (VC), C_2H_4 and Cl_2 gas first enters a reactor at 90°C and 1.5 atm, to which C_2H_4 , Cl_2 , VC, and HCL are produced. The pressure of the product stream is then increased to 26 atm by a pump before it enters a gas furnace, where it is heated up, and exits at 500°C and 26 atm. The stream then passes through a series of cooler, condenser, and a throttling valve, reaching a temperature of 6°C and a pressure of 12 atm before entering the first distillation column. As the stream goes through the column, HCL is isolated to the distillate section and goes through a condenser to be cooled down to -26.2°C using a refrigerant before passing through a reflux drum. The HCL is then partially purged through a distillate stream, and the remaining HCL is recycled back to the top stage of the column via a reflux stream. With HCL going to the distillate section, VC and C_2H_4 , Cl_2 go to the bottom section where they pass through a reboiler to be heated up to 93°C using steam and recycled back to the bottom stage of the column before passing to a throttling valve to reduce the stream's pressure down to 4.8 atm. The stream now enters a second distillation column at 93°C and 4.8 atm. The VC and C_2H_4 , Cl_2 is then separated, with our VC product going to the top distillate section to be cooled down to 33°C using cooling water before passing through a reflux drum and collected as the desired product. Some of the VC is also refluxed back into the top stage of the column. The C_2H_4 , Cl_2 goes to the bottom of the column, where it passes through a reboiler to be heated to 146°C using steam before being refluxed to the bottom stage of the column. C_2H_4 , Cl_2 stream then goes through a throttling valve to reduce the pressure to 4.8 atm before being recycled back to the outlet of the reactor at 146°C and 4.8 atm.

In this particular process in producing VC, there are 2 major unit operations. The first one is heating, which happens at the furnace that heats the product stream of the reactor to 500°C using natural gas. And the second is separation, which happens at both distillation columns. The first column separates HCL from VC and C_2H_4 , Cl_2 , and the second distillation column separates VC and C_2H_4 , Cl_2 .

```

graph LR
    Feed1[C2H4  
Cl2  
243,146 lb/hr] --> Reactor
    subgraph Reactor [Reactor]
        direction TB
        In1[C2H4  
Cl2]
        Out1[65% Conversion]
    end
    Reactor -- "HCL  
VC  
C2H4Cl2  
158,045 lb/hr" --> Furnace
    subgraph Furnace [Furnace]
        direction TB
        In2[C2H4Cl2]
        Out2[244,602 lb/hr]
    end
    Furnace -- "HCL  
VC  
C2H4Cl2" --> D1
    subgraph D1 [1st Distillation Column]
        direction TB
        In3[C2H4Cl2]
        Out3[58,045 lb/hr]
        Out4[VC  
C2H4Cl2  
188,557 lb/hr]
    end
    D1 -- "VC  
C2H4Cl2  
100,000 lb/hr" --> D2
    subgraph D2 [2nd Distillation Column]
        direction TB
        In4[C2H4Cl2]
        Out5[VC  
C2H4Cl2]
    end
    D2 -- "C2H4Cl2  
86,557 lb/hr" --> Reactor

```

The diagram illustrates the production of vinyl chloride (VC) from ethylene (C₂H₄) and dichloroethane (C₂H₄Cl₂). The process involves a reactor, a furnace, and two distillation columns. Feed rates and conversion percentages are specified for each unit.

- Feed 1:** C₂H₄ and Cl₂ enter the Reactor at a rate of 243,146 lb/hr.
- Reactor:** The feed is converted by 65%. The output is a mixture of HCL, VC, and C₂H₄Cl₂ at a rate of 158,045 lb/hr.
- Furnace:** The mixture from the reactor enters the furnace. The output is a mixture of HCL, VC, and C₂H₄Cl₂ at a rate of 244,602 lb/hr.
- 1st Distillation Column:** The mixture from the furnace enters the column. The column separates the mixture into two streams:
 - Top Product:** HCL, VC, and C₂H₄Cl₂ at a rate of 58,045 lb/hr.
 - Bottom Product:** VC, C₂H₄Cl₂, and HCL at a rate of 188,557 lb/hr.
- 2nd Distillation Column:** The bottom product from the 1st column enters the 2nd column. The column separates the mixture into two streams:
 - Top Product:** VC, C₂H₄Cl₂, and HCL at a rate of 100,000 lb/hr.
 - Bottom Product:** C₂H₄Cl₂ at a rate of 86,557 lb/hr.
- Recycle:** The bottom product from the 2nd column (C₂H₄Cl₂) is recycled back to the Reactor.

Figure 1: BFD with Species Mass Flow

[illegible]

Figure 2: Full PFD of Production of Vinyl Chloride

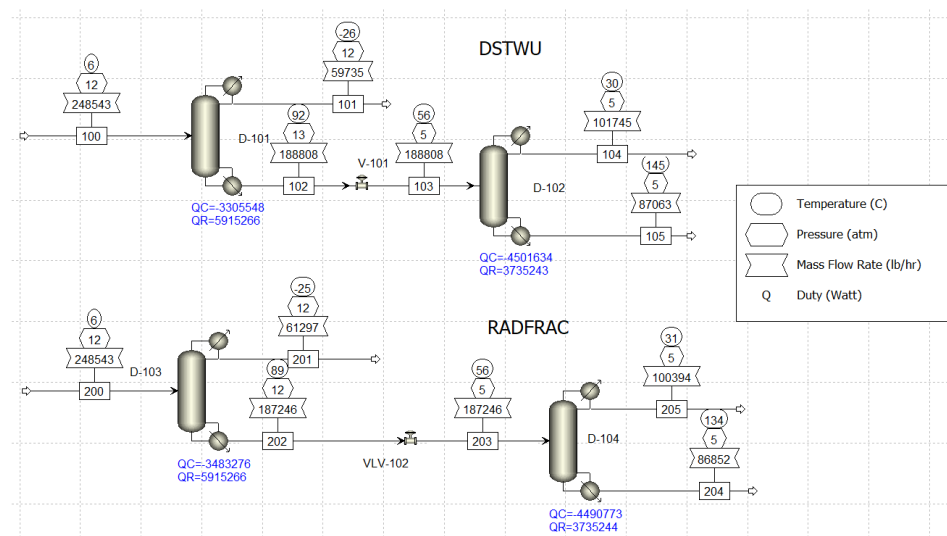
Distillation Column Simulation

Figure 3: ASPEN Simulation of Distillation Columns with DSTWU and RADFRAC Models

1 st Column Reflux Ratio	0.17
1 st Column Number of Stages	26
1 st Column Feed Location	13
2 nd Column Reflux Ratio	0.11
2 nd Column Number of Stages	33
2 nd Column Feed Location	15
DSTWU VC Feed Comp	0.39
DSTWU HCL Feed Comp	0.39
DSTWU EDC Feed Comp	0.21
RADFRAC VC Feed Comp	0.39
RADFRAC HCL Feed Comp	0.39
RADFRAC EDC Feed Comp	0.21
DSTWU VC Product Comp (Purity)	0.99
RADFRAC VC Product Comp (Purity)	0.98

Table 1: Summary of Reflux Ratio, Number of Stages, Feed Comp, and Purity

Design Considerations and Trade-Offs

In the making of the DSTWU model, the first column was assumed to have an operating condition of 6°C and 12 atm. It was also assumed that Vinyl Chloride (VC) and Hydrochloric (HCL) are equimolar. The reflux ratio within DSTWU is set as -1.15 since the number of stages is unknown currently. HCL was then set as the light key with a recovery rate of 99.5% and VC was set as the heavy key in the distillate section with a recovery rate of 0.5%. The condenser had

a pressure of 12 atm, and the reboiler had a pressure of 13 atm since it was assumed that there was a bit of vapor within the reboiler due to the solution boiling. The second column within the DSTWU model had similar assumptions compared to the first column, with the reflux ratio as - 1.15, VC being the light key in the distillate with a recovery rate of 99.5% and Dichloroethane (EDC) being the heavy key in the distillate with a recovery rate of 0.5%. The condenser in the second column had a lower pressure of 4.8 atm due to the throttling valve and the reboiler had a pressure of 5.2 atm. It was also assumed that the condenser and reboiler are well-insulated.

For processes that involve distillation columns, a higher purity within the product means we would need more stages, leading to a larger condenser/reboiler and ultimately a higher level of energy consumption. The addition of more stages would affect the measurements of the column, specifically the height and in turn drastically increase the capital cost of the plant. Moreover, other effects, such as an increased reflux ratio, would make the diameter of the column larger, which directly affects the capital cost. The diameter increase isn't the only thing impacting capital cost, but an increase in reflux ratio also increased the energy consumption of the condenser and reboiler, which in turn inflates both the capital cost and utility cost.

AI Prompt and Output Appendix

Kendall: CoPilot AI

Prompt

Task 2: You are employed as a process engineering consultant at a chemical engineering design firm. Your team has been assigned to develop a preliminary design report for a new vinyl chloride monomer (VC) production facility to be built in Texas, USA. The planned plant will have a production capacity of 100,000 lb/hr of VC and will supply downstream polyvinyl chloride (PVC) manufacturing operations. Your task is to prepare a professional introductory design report that demonstrates your understanding of the process, simulates key separation units, and critically evaluates how AI tools can support — and fail — in process design decision-making. Your work must address the following: Draft process descriptions and PFD outlines • Suggest distillation column design parameters (stage count, reflux ratio, operating pressure) • Propose operating conditions or alternative column configurations • Suggest optimizations for energy integration or column sequencing • Complete prompt and output log (unedited)

Result

Here's a professional introductory design report tailored for the vinyl chloride monomer (VCM) production facility in Texas:

Preliminary Design Report: Vinyl Chloride Monomer (VCM) Production Facility – Texas, USA

1. Introduction

This report outlines the preliminary design for a VCM production plant with a capacity of 100,000 lb/hr, supplying downstream PVC manufacturing. The facility will be located in Texas, leveraging proximity to petrochemical infrastructure and feedstock availability. The design focuses on process understanding, simulation of key separation units, and a critical evaluation of AI tools in process engineering.

2. Process Description

2.1 Overview

VCM is primarily produced via thermal cracking of ethylene dichloride (EDC). The process involves:

- **EDC Cracking:** EDC is vaporized and thermally cracked at high temperatures (~500–550°C) to yield VCM, HCl, and unreacted EDC.
- **Quenching:** The cracked gas is rapidly cooled to prevent further reactions.
- **Separation:** A series of distillation columns separate VCM, HCl, and recycle EDC.

2.2 Process Flow Diagram (PFD) Outline

EDC Feed → Vaporizer → Cracking Furnace → Quench Tower → HCl Removal → VCM/EDC Separation → VCM Product → PVC Plant

- **Unit Operations:**
 - Cracking Furnace
 - Quench Tower
 - Absorber for HCl removal
 - Distillation Column for VCM/EDC separation
 - Heat exchangers for energy recovery

Sources:

3. Distillation Column Design Parameters

3.1 VCM/EDC Separation Column

- **Stage Count:** 25–35 theoretical stages
- **Reflux Ratio:** 2.5–3.5 (optimized for purity and energy)
- **Operating Pressure:** ~2–3 atm to balance volatility and equipment cost
- **Column Type:** Tray column with sieve trays or structured packing for high separation efficiency

3.2 HCl Removal Column

- **Stage Count:** ~10–15 stages
- **Reflux Ratio:** ~1.5–2.0
- **Operating Pressure:** Slightly above atmospheric to minimize corrosion and material degradation

Sources:

4. Operating Conditions & Alternative Configurations

4.1 Cracking Furnace

- **Temperature:** 500–550°C
- **Residence Time:** ~10–20 seconds
- **Conversion:** ~50–60% per pass

4.2 Alternative Configurations

- **Reactive Distillation:** Potential for simultaneous cracking and separation
- **Membrane Separation:** Emerging tech for HCl recovery
- **Pressure Swing Distillation:** For energy-efficient separation of azeotropes

5. Energy Integration & Column Sequencing

5.1 Energy Optimization

- **Heat Integration:** Use cracked gas heat to preheat EDC feed

- **Pinch Analysis:** Identify minimum utility consumption
- **Steam Generation:** Recover heat from furnace effluent

5.2 Column Sequencing

- **Sequence:** HCl removal → VCM/EDC separation
- **Justification:** Prevent HCl corrosion in downstream units and improve VCM purity

Ify: Gemini Ai Pro

Prompt

Task 2: You are employed as a process engineering consultant at a chemical engineering design firm. Your team has been assigned to develop a preliminary design report for a new vinyl chloride monomer (VC) production facility to be built in Texas, USA. The planned plant will have a production capacity of 100,000 lb/hr of VC and will supply downstream polyvinyl chloride (PVC) manufacturing operations. Your task is to prepare a professional introductory design report that demonstrates your understanding of the process, simulates key separation units, and critically evaluates how AI tools can support — and fail — in process design decisionmaking. Your work must address the following: Draft process descriptions and PFD outlines

- Suggest distillation column design parameters (stage count, reflux ratio, operating pressure)
- Propose operating conditions or alternative column configurations
- Suggest optimizations for energy integration or column sequencing
- Complete prompt and output log (unedited)

Result

Of course. Here is a preliminary design report for the proposed vinyl chloride monomer production facility.

MEMORANDUM**To:** Project Stakeholders**From:** Gemini Engineering Consultants**Date:** October 16, 2025**Subject:** Preliminary Design Report: 100,000 lb/hr Vinyl Chloride Monomer Production Facility**1.0 Introduction**

This report outlines a preliminary process design for a new grassroots vinyl chloride (VC) monomer production facility to be located in Texas, USA. The plant is designed for a nominal production capacity of 100,000 lb/hr (approx. 45.4 metric tons/hr) of polymer-grade vinyl chloride. The product will serve as feedstock for downstream polyvinyl chloride (PVC) manufacturing operations. This document covers the recommended process technology, a description of the major process units, key separation design parameters, and an evaluation of the role of artificial intelligence (AI) in the design process.

2.0 Process Technology Selection and Description

The recommended technology for this facility is the **balanced process**, which integrates the direct chlorination and oxychlorination of ethylene to produce the intermediate 1,2-dichloroethane (EDC). The EDC is subsequently cracked via thermal pyrolysis to yield vinyl chloride and hydrogen chloride (HCl). This process is the industry standard due to its high efficiency and atomic economy, as the HCl byproduct from the cracking step is recycled and consumed in the oxychlorination step.

The overall process can be divided into four primary sections:

1. **EDC Synthesis:** Ethylene is reacted with chlorine (direct chlorination) and with HCl and oxygen (oxychlorination) to form EDC.
2. **EDC Purification:** Crude EDC from the synthesis section is purified via washing and distillation.
3. **EDC Cracking:** Purified EDC is thermally cracked to produce VC and HCl.
4. **VC Purification:** The cracker effluent is separated to yield high-purity VC product, with unreacted EDC and HCl being recycled.

2.1 Process Flow Diagram (PFD) Outline

- **R-101 (Direct Chlorination Reactor):** Ethylene and chlorine react in a liquid EDC medium with a ferric chloride (FeCl_3) catalyst. $\text{C}_2\text{H}_4 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_4\text{Cl}_2 + \text{Heat}$
- **R-102 (Oxychlorination Reactor):** Ethylene, recycled HCl, and air/oxygen react over a copper chloride (CuCl_2) catalyst in a fluidized bed or fixed bed reactor. $\text{C}_2\text{H}_4 + 2\text{HCl} + \frac{1}{2}\text{O}_2 \rightarrow \text{C}_2\text{H}_4\text{Cl}_2 + \text{H}_2\text{O} + \text{Heat}$
- **T-101 (EDC Purification Train):** A series of distillation columns designed to remove light ends (e.g., ethyl chloride) and heavy ends (e.g., higher chlorinated hydrocarbons) from the crude EDC stream before it is sent to cracking.
- **F-101 (Cracking Furnace):** Purified liquid EDC is vaporized and heated to approximately 500°C for pyrolysis. $\text{C}_2\text{H}_4\text{Cl}_2 + \text{Heat} \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{HCl}$
- **T-102 (HCl Column):** Separates HCl from the cracker outlet stream for recycling to R-102.
- **T-103 (VC Column):** The final purification column, which separates the VC product from unreacted EDC. The EDC is recycled back to the EDC purification train.

```
Pythonimport sys
from rdkit import Chem
from rdkit.Chem import Draw

# Create a list for molecules and their legends
mols = []
legends = []

# Add molecules and legends to the lists
smiles_list = {
    'Ethylene (Reactant)': 'C=C',
    'Chlorine (Reactant)': 'ClCl',
    '1,2-Dichloroethane (Intermediate)': 'C(Cl)CCl',
    'Vinyl Chloride (Product)': 'C=CCl'
```

```
}

for name, smiles in smiles_list.items():
    mol = Chem.MolFromSmiles(smiles)
    if mol:
        mols.append(mol)
        legends.append(name)

# Generate the grid image
if mols:
    img = Draw.MolsToGridImage(
        mols,
        molsPerRow=4,
        subImgSize=(300, 300),
        legends=legends
    )
    img.save('molecules.png')
else:
    # Create a blank image or handle the error if no molecules were processed
    from PIL import Image
    img = Image.new('RGB', (100, 30), color = 'white')
    img.save('molecules.png')
```



3.0 Distillation Column Design Parameters

The purification section is critical for meeting product specifications and ensuring efficient recycling of reactants. Below are preliminary design suggestions for the two primary distillation columns in the VC purification train.

3.1 HCl Column (T-102)

- **Service:** Separates HCl (light key) from VC and unreacted EDC (heavy keys).
- **Operating Pressure:** A higher pressure is recommended, in the range of **10-15 bar**. This raises the boiling point of HCl (Normal BP: -85°C) to a level where it can be condensed using standard refrigerants or, potentially, chilled water, reducing operational costs for the condenser.
- **Stage Count:** An estimated **20-30 theoretical stages** would be a reasonable starting point for simulation. This provides a balance between capital cost (column height) and energy cost (reflux ratio).
- **Reflux Ratio:** A reflux ratio of **1.3 times the minimum reflux ratio ($SR_{\min}$)** is proposed. This value offers a good trade-off, moving away from the infinite energy demand at $SR_{\min}$ and the infinite capital cost at minimum stages.

3.2 VC Column (T-103)

- **Service:** Separates the final vinyl chloride product (light key) from unreacted EDC (heavy key). This column's performance is critical for product purity.
- **Operating Pressure:** Suggested operating pressure is **4-6 bar**. This pressure is sufficient to condense the VC overhead product with standard cooling water ($\sim 30\text{-}40^{\circ}\text{C}$) while avoiding excessively high temperatures in the reboiler, which could lead to fouling or unwanted side reactions.
- **Stage Count:** Due to the relatively close boiling points of VC (-13.4°C) and EDC (83.5°C), a higher number of stages is required for sharp separation. A preliminary estimate is **50-70 theoretical stages**.
- **Reflux Ratio:** A reflux ratio of **1.2 times the minimum reflux ratio ($SR_{\min}$)** is recommended to ensure high product purity ($>99.9\%$) without excessive energy consumption.

4.0 Optimization and Alternative Configurations

Significant opportunities exist for process optimization, particularly concerning energy integration.

- **Energy Integration:** The process involves highly exothermic reactions (Direct Chlorination, Oxychlorination) and energy-intensive unit operations (EDC Cracking, Distillation Reboilers). A **pinch analysis** should be conducted during detailed design to develop a heat exchanger network. Heat from the reactor outlets can be used to generate medium-pressure steam, which can then be used to drive the reboilers for T-101, T-102, and T-103. The hot effluent from the cracking furnace (F-101) should be used to preheat the furnace's EDC feed, recovering a significant amount of high-grade heat.
- **Dividing Wall Column (DWC):** For the EDC purification train (T-101), which involves removing both light and heavy impurities, a DWC could be considered. This configuration can perform the task of two separate columns within a single shell, offering substantial savings in both capital expenditure and energy consumption.

Hussein: ChatGPT 5

Prompt:

Task 2: You are employed as a process engineering consultant at a chemical engineering design firm. Your team has been assigned to develop a preliminary design report for a new vinyl chloride monomer (VC) production facility to be built in Texas, USA. The planned plant will have a production capacity of 100,000 lb/hr of VC and will supply downstream polyvinyl chloride (PVC) manufacturing operations. Your task is to prepare a professional introductory design report that demonstrates your understanding of the process, simulates key separation units, and critically evaluates how AI tools can support — and fail — in process design decisionmaking. Your work must address the following:

Phase 1 – Individual AI Exploration

Each student must individually select one AI tool (e.g., ChatGPT, Gemini, Claude, Copilot, Perplexity, etc.) and use it to attempt solving the distillation and process design problem. The AI can be used to:

- Draft process descriptions and PFD outlines
- Suggest distillation column design parameters (stage count, reflux ratio, operating pressure)
- Propose operating conditions or alternative column configurations

- Suggest optimizations for energy integration or column sequencing
- Complete prompt and output log (unedited)

Results

Vinyl Chloride Monomer (VCM) Facility – Introductory Design Report

Plant Location: Texas, USA

Product Rate (Basis): 100,000 lb/hr VCM (on-spec, polymer-grade)

Prepared by: Process Engineering Consultant (Preliminary Design)

Date: Oct 16, 2025

1) Executive Summary

This report outlines a preliminary design for a world-scale vinyl chloride monomer (VCM) facility integrated with downstream PVC operations. The selected technology path is the conventional **EDC thermal cracking** route with **EDC synthesis via direct chlorination and oxychlorination of ethylene**. We provide draft process descriptions and PFD outlines; suggest distillation column parameters (stage counts, reflux ratios, pressures); propose operating conditions and configuration alternatives; and highlight energy-integration opportunities (heat recovery, MVR, side reboilers). We also append an **AI prompt & output log** and discuss where AI aids or fails in process design.

High-level basis & sizing (order-of-magnitude):

- VCM production: **100,000 lb/hr** → **~1,600 lbmol/hr** (MW VCM ≈ 62.5 lb/lbmol)
- Stoichiometry ($\text{EDC} \rightarrow \text{VCM} + \text{HCl}$): $1:1:1 \Rightarrow$ co-product **HCl ~1,600 lbmol/hr $\approx$ 58,300 lb/hr**
- Net EDC consumption (to product): **~1,600 lbmol/hr $\approx$ 158,300 lb/hr** (fresh makeup $\approx$ consumption; unreacted EDC is recycled)
- EDC cracker single-pass conversion (design target): **55–65%**; selectivity to VCM $\geq 99\%$ (by preliminary assumption)
- Key separations: **HCl removal & recovery, VCM/EDC splitter, lights/byproduct removal, EDC purification for recycle**

Key column targets (initial):

- HCl Removal Column: 20–28 stages, **$R \approx 1.2\text{--}1.6 \times R_{\min}$** , **$P = 16\text{--}20$ barg** to condense VCM/EDC while taking HCl overhead.

- VCM/EDC Splitter: 40–60 stages, **R $\approx$ 2.0–3.0, P = 8–10 barg** (overhead VCM easily condensed with cooling water; minimize polymerization risk).
- Lights/VCM Column (optional, upstream of VCM/EDC): 18–26 stages, **R $\approx$ 1.5–2.2, P = 12–16 barg** for benign condensation.
- EDC Purifier/Deheavies: 24–36 stages, **R $\approx$ 1.5–2.5, near-atm to 2 barg** (optimize vs fouling/stability).

Energy integration highlights: cracked-gas heat recovery to MP steam; **feed–effluent exchanger** around EDC cracker coils; **MVR** on VCM overhead; **heat cascade** from high-P HCl condenser to reboilers; side-reboilers and inter-condensers where economic.

2) Process Description & Block/PFD Outlines

2.1 Technology Route (Overview)

1. EDC Synthesis:

- **Direct chlorination (DC):** Ethylene + Cl₂ → EDC (exothermic).
- **Oxychlorination (OC):** Ethylene + HCl + O₂ → EDC + H₂O (CuCl₂/Al₂O₃ or similar catalyst). Uses crack-derived HCl as feedstock.
- Combined DC/OC loop maintains EDC inventory and minimizes net Cl₂ use.

2. EDC Purification:

- Remove light ends and heavies; deliver **on-spec EDC** to the cracking furnaces.

3. EDC Thermal Cracking:

- **EDC → VCM + HCl** (endothermic, gas-phase). Furnace coils (e.g., Incoloy) at **480–520 °C, 15–30 psig** coil outlet; short residence (seconds) to maximize selectivity, minimize tar/coke.
- Effluent is **rapidly quenched** (EDC cold quench or oil quench) to arrest side reactions.

4. HCl Recovery & Drying:

- Cracked gas (VCM/EDC/HCl/lights) contacted/partially condensed; **HCl is separated/recovered** (dry or absorbed and re-stripped) to return to **oxychlorination**. Dehydration to protect catalysts.

5. VCM Purification:

- **VCM/EDC distillation and lights removal** to meet polymer-grade specs (e.g., < ppm heavies, water, acetylene, chlorinated byproducts). **Polymerization inhibitor** (e.g., p-tert-butylcatechol) dosed as needed.

6. EDC Recycle:

- Unreacted EDC recovered and recycled to EDC cracker feed; purged to heavies/SMPO to control chlorinated heavies.

7. Utilities & Ancillaries:

- **Caustic/amine scrubbers** for vent treatment, **N₂ blanketing**, **inhibitor dosing**, **flare**, **wastewater neutralization**, **cooling water**, **steam/power**, **instrument air**, **dryers** (molecular sieves if needed).

2.2 Draft PFD Outlines (Textual)

Block 100 – EDC Synthesis & Purification

- B101 DC Reactor → C101 DC Cooler/Condenser → D101 DC Separator → T101 EDC Finisher (lights/deheavies)
- B102 OC Reactor + Air Separation/O₂ Feed → E101 OC Waste-Heat Boiler → D102 Washer/Separator → T102 EDC Purifier

Block 200 – Cracking & Quench

- H201 Cracking Furnace (coils) → E201 Feed–Effluent Exchanger → Q201 Quench Tower (EDC recycle) → D201 Phase Separator (HCl-rich gas / VCM-EDC-rich liquid)

Block 300 – HCl Recovery & Drying

- T301 **HCl Removal Column** (HCl overhead) → C301 HCl Condenser/Absorber → B301 HCl Dryer → Return HCl to OC
- Bottoms (VCM/EDC + lights) → D301 Knockout/Drum → T302 **Lights/VCM Column** (optional, if lights build-up is material)

Block 400 – VCM/EDC Purification & Recycle

- T401 **VCM/EDC Splitter** (VCM overhead to product; EDC bottoms to recycle)
- C401/C402 Overhead condensation & reflux system with inhibitor injection
- P401 VCM Product Compressor/Blower (if pressurization needed) → S401 Dryers/Polishing → TK401 Product Storage (refrigerated/pressurized)

Block 500 – Heavies Management

- T501 Deheavies/Stripper (chlorinated heavies purge) → Incineration/SMPO or appropriate treatment and safe disposal

Block 600 – Utilities & OSBL

- Steam gens from crack effluent heat recovery; cooling water systems; flare; nitrogen system; DCS/SIS; electrical; wastewater; firewater/foam.

3) Property Methods & Simulation Approach (Prelim)

- **VCM/EDC/HCl/lights system:** Use **Peng–Robinson (PR) with volume translation** or **PR-BM** for VLE at moderate pressure; include **BIPs** from reliable databanks. For VCM polymerization risk, maintain inhibitor modeling (pseudo-component) as needed.
- **Aqueous HCl/OC section:** **Electrolyte NRTL** for absorption/stripping and OC wash water systems; include dissociation equilibria.
- **Sensitivity:** Check SRK-BM vs PR-BM for nonideal HCl partitioning; compare with **NRTL-HOC** for chlorinated organics if data suggests non-ideal vapor-phase association.

Simulation deliverables (suggested):

- Converged distillation blocks for **T301 HCl Removal** and **T401 VCM/EDC Splitter** with shortcut (Winn–Underwood–Gilliland) then rigorous RadFrac equivalents.
- Heat curves and **ΔT_{min} targeting** for heat integration.

4) Preliminary Material Balance (Order-of-Magnitude)

Basis: 100,000 lb/hr VCM product.

- VCM MW ≈ 62.5 lb/lbmol → **1,600 lbmol/hr VCM**.
- Stoich: $\text{EDC} \rightarrow \text{VCM} + \text{HCl} \Rightarrow \text{HCl} \sim 1,600 \text{ lbmol/hr}$ (**$\sim 58,300$ lb/hr**).
- Net EDC consumed: **$\sim 1,600$ lbmol/hr** (**$\sim 158,300$ lb/hr**); fresh EDC makeup equals consumption after recycle.
- Lights/byproducts assumed $< 1\text{--}3$ mol% of cracked gas (tuned during detailed simulation); heavies purge $0.5\text{--}1$ wt% of EDC circulation to manage buildup.

5) Key Separation Design – Shortcut Estimates & Initial Specs

Use **Fenske–Underwood–Gilliland (FUG)** for initial designs; refine with rigorous column models.

5.1 T301 – HCl Removal Column (Acid Gas Overhead)

- **Duty/Role:** Remove HCl overhead from VCM/EDC mixture to protect downstream distillations and recover HCl for OC.
- **Operating Pressure:** **16–20 barg** to ensure VCM/EDC condense in the overhead system while HCl remains light; facilitates cooling-water condensation and robust pressure margin.
- **Number of Stages:** **20–28** (includes reboiler & condenser) – refine after VLE fit; higher pressure eases condensation but increases relative volatility coupling.
- **Reflux Ratio:** Target $R \approx 1.2\text{--}1.6 \times R_{\min}$ (start at $R = 1.8\text{--}2.5$ actual) to balance capital/energy; tighten with U/G correlations.
- **Internals:** Structured packing (acid-resistant) or trays (e.g., valve trays) with **HCl-resistant metallurgy**; apply **chloride stress corrosion** controls; design for inhibitor carryover control.
- **Overhead System:** Partial condenser + HCl compression/absorption path; optional **dry** HCl route to OC after dehydration.

5.2 T401 – VCM/EDC Splitter (Product Purification)

- **Duty/Role:** Sharp split: **VCM overhead to product, EDC bottoms to recycle.**
- **Operating Pressure:** **8–10 barg** – keeps VCM condensation feasible with CW; mitigates refrigeration needs; reduces polymerization risk vs too-low pressure.
- **Number of Stages:** **40–60** (initial target **~50 stages**) depending on required VCM purity (polymer-grade) and allowable EDC in overhead (ppm-level) and VCM loss in bottoms.
- **Reflux Ratio:** $R \approx 2.0\text{--}3.0$ (aim mid-range **~2.5**); validate via shortcut design with key components $\alpha(\text{VCM/EDC})$ at selected P.
- **Internals:** High-capacity **structured packing** to limit ΔP (polymerization risk rises with hot holdup); consider **dual feed** if lights are present.
- **Overhead System:** Condenser with **inhibitor injection** and **N₂ blanketing**; receiver with cold return reflux; **polishing dryer** before storage.

5.3 Optional – Lights/VCM Column (T302)

- **Duty/Role:** Remove light impurities (e.g., uncondensables, very light chlorinateds) before the VCM/EDC splitter to protect α and tray counts.

- **Operating Pressure:** 12–16 barg.
- **Number of Stages:** 18–26.
- **Reflux Ratio:** 1.5–2.2.

5.4 EDC Purifier/Deheavies (T102/T501)

- **Duty/Role:** Polish EDC before recycle to the furnaces; purge heavies/byproducts to avoid fouling.
- **Operating Pressure:** ~1–2 barg (near-atm to simplify heat duties).
- **Number of Stages:** 24–36.
- **Reflux Ratio:** 1.5–2.5.

Control & Safeguards (all columns):

- Differential pressure monitors; anti-polymerant dosing; oxygen exclusion; SIS trips on high ΔP /high T; reliefs to flare.

6) Operating Conditions & Alternatives

Base Cracker Conditions: Furnace outlet 480–520 °C, residence 1–3 s, coil outlet pressure 15–30 psig; quench to <200 °C within seconds, then to <50 °C before compression/absorption.

Column Alternatives & Enhancements:

- **High-pressure HCl Column** to enable **heat cascade**: use its **hot overhead condenser** as a heat source for an EDC/VCM reboiler via an **intermediate loop** (pinch-driven integration).
- **MVR on VCM Overhead**: Compress a slipstream of VCM overhead vapor to supply reboiler duty (reduce steam). Check economics vs electricity cost.
- **Dividing Wall Column (DWC)** options for lights/VCM/EDC ternary separation if impurity spectra justify (CAPEX vs OPEX trade).
- **Two-pressure VCM splitters** (prefractionator + main) to minimize reflux and condenser duties; or **side reboiler/side condenser** to match composite curves.
- **Reactive absorption** of HCl into water to produce **20–32 wt% HCl** product; add stripper to recycle dry HCl to OC (choose based on OC mass balance).

7) Energy Integration & Pinch Opportunities

- **Cracked gas heat recovery** → Generate **MP steam** (e.g., 8–13 barg) in waste-heat boilers pre-quench; ensure fast quench to protect selectivity.
- **Feed-effluent exchanger (FEE)** around cracker: preheat EDC feed with hot effluent while respecting $\Delta T_{\min} \sim 15\text{--}25\text{ }^{\circ}\text{C}$ to control fouling/coking risk.
- **Column heat cascade**: Couple **T301 condenser** (hot at high P) to **T401 reboiler** (cooler), via closed-loop heat transfer oil or tempered water.
- **MVR/TVR** evaluation for VCM splitter; **C3+ refrigeration** may be avoided at **8–10 barg** if cooling water is sufficient.
- **Flash integration**: Use **let-down flash** of high-P VCM to generate refrigeration for product subcooling.

8) Safety, Materials, and Environmental Notes (Prelim)

- **Toxicity & Flammability**: VCM is carcinogenic and flammable; HCl is corrosive/toxic. Enforce **closed systems, double mechanical seals, N₂ blanketing, detectors (HCl/VCM)**.
- **Polymerization control**: Inhibitors (e.g., TBC) in VCM circuits; minimize hot liquid holdup; low ΔP packing; avoid oxygen ingress.
- **Materials**: Alloyed steels/Ni-based for cracker coils; **rubber-lined/FRP** for acid services; 316L/Alloy 20 where compatible; chloride SCC checks.
- **Environmental**: Vent scrubbing (caustic/amine); flare system; brine/wastewater neutralization; OC catalyst and chloride handling.

9) Roadmap to Rigorous Simulation

1. **Thermo selection & data regression** for VCM/EDC/HCl; validate K-values vs literature.
2. **Shortcut distillation (FUG)** → **RadFrac** with sensitivity on **P, R, Nstages**; size condensers/reboilers; check **HETP/tray efficiencies**.
3. **Heat integration**: Composite curves; **$\Delta T_{\min}$ targeting**; evaluate MVR/TVR.
4. **Hydraulics**: Flooding checks, capacity factor, downcomer backup; select packing/tray vendors.

5. **Controls:** Pressure control with condenser duty and vent; reflux/boilup ratio loops; product purity analyzers.

Basil: Claude

Prompt:

Given this question, can you provide me with some ideas of parameters to set for the dimensions of the column as well as some ideas for optimization? (submitted along with a photo of the problem description given in the assignment).

Results:

Process Design Parameters & Optimization Ideas for VC Distillation Column

I'll help you develop a comprehensive approach for the distillation column design and optimization strategy for this vinyl chloride production facility.

Key Column Design Parameters

1. Operating Conditions

- **Operating Pressure:** 5-15 bar (VC boiling point: -13.4°C at 1 atm, need elevated pressure)
- **Temperature Range:**
 - Top: $10\text{-}30^{\circ}\text{C}$ (condenser)
 - Bottom: $50\text{-}80^{\circ}\text{C}$ (reboiler)
- **Reflux Ratio:** 1.5-3.0 (initial estimate, to be optimized)
- **Feed Location:** 40-60% from bottom (depends on feed quality)

2. Column Dimensions

Diameter Calculation Basis:

- Vapor flow rate: $\sim 45,000$ lb/hr (assuming 45% overhead)
- Liquid flow rate: $\sim 55,000$ lb/hr (bottoms)
- Vapor velocity: 70-80% of flooding velocity
- **Estimated Diameter:** 6-10 feet (1.8-3.0 m)

Height Parameters:

- **Number of Theoretical Stages:** 30-50 stages (depends on separation requirement)

- **Actual Trays:** 40-70 trays (with 60-75% tray efficiency)
- **Tray Spacing:** 18-24 inches (0.45-0.6 m)
- **Total Column Height:** 60-120 feet (18-36 m)
- **Additional Height:**
 - Top disengagement: 5-8 feet
 - Bottom surge: 8-12 feet
 - **Total Height:** 75-140 feet

3. Internals Selection

Tray Type Options:

- **Sieve Trays:** Lower cost, good for clean service
- **Valve Trays:** Better turndown ratio (recommended for 100,000 lb/hr capacity)
- **Bubble Cap Trays:** Higher efficiency but more expensive

Packing Alternative:

- Structured packing (e.g., Mellapak, Flexipac) for:
 - Lower pressure drop
 - Higher capacity
 - Better for vacuum operations if needed

Optimization Strategy

1. Economic Optimization (Primary Objective)

Total Annual Cost (TAC) Minimization:

$TAC = \text{Capital Cost (Annualized)} + \text{Operating Cost}$

Key Variables to Optimize:

- **Reflux Ratio** (most sensitive parameter)
- **Number of Stages (N)**
- **Feed Stage Location (NF)**
- **Column Pressure**

Trade-offs:

Lower Reflux Ratio → Smaller reboiler/condenser → Lower capital cost

Higher Reflux Ratio → More stages needed → Lower operating cost

2. Optimization Parameters & Ranges

Parameter	Typical Range	Optimization Impact
Reflux Ratio (R/R_{min})	$1.2 - 2.0 \times R_{min}$	High - affects energy consumption
Number of Stages	20 - 80	Medium - capital vs. separation
Feed Stage Location	$0.3 - 0.7 \times N$	Medium - affects product purity
Operating Pressure	5 - 15 bar	High - affects utilities & safety
Column Diameter	6 - 12 ft	Medium - affects capital cost

3. Multi-Objective Optimization Approach

Objective Functions:

1. **Minimize TAC** (\$/year)
2. **Maximize Product Purity** (>99.9% VC)
3. **Minimize Energy Consumption** (kW)
4. **Maximize Safety Index** (minimize hazardous inventory)

Constraints:

- VC purity $\geq 99.5\%$ (PVC manufacturing requirement)
- Pressure < 15 bar (safety/metallurgy limits)
- Diameter < 12 ft (manufacturing/transport limits)
- ΔP per tray < 0.15 psi (flooding prevention)

4. Recommended Optimization Workflow

Step 1: Initial Design (AI-Assisted)

- Use Aspen Plus with heuristics
- Run sensitivity analysis on reflux ratio ($1.2-2.5 \times R_{min}$)
- GA optimization for N and NF

Step 2: Detailed Simulation (Human + AI)

- Rigorous tray rating calculations
- Hydraulic design (flooding, weeping, entrainment)
- Dynamic simulation for control system design

Step 3: Economic Analysis (Human-Led)

- Capital cost estimation (CAPCOST, vendor quotes)
- Operating cost (utilities: steam at \$8/1000 lb, cooling water at \$0.05/1000 gal)
- NPV analysis with 10-year horizon

Step 4: Safety & Operability Review (Human-Critical)

- HAZOP study
- Layer of Protection Analysis (LOPA)
- Operating procedures development
- Emergency shutdown system design

Step 5: Final Optimization (Hybrid)

- AI fine-tunes within safe operating envelope
- Human validates against regulations and best practices

Specific Recommendations for Your VC Column**Starting Point Design:**

- **Diameter:** 8 feet (allow for future expansion)
- **Height:** 100 feet (50 actual trays, 24-inch spacing)
- **Reflux Ratio:** 2.0 (conservative for startup)
- **Pressure:** 10 bar (balance utilities and safety)
- **Material:** SS316L (chloride service)
- **Internals:** Valve trays with 18-inch spacing

Optimization Focus Areas:

1. **Energy Integration:**
 - Use VC overhead vapor to preheat feed
 - Integrate with ethylene dichloride (EDC) cracking furnace

2. Pressure Optimization:

- Higher pressure → easier condensation (cooling water vs. refrigeration)
- Lower pressure → easier reboiling (lower steam pressure)
- **Sweet spot:** 8-12 bar

3. Tray Efficiency Enhancement:

- Add inlet weirs to improve liquid distribution
- Use high-capacity valve trays for 20% turndown capability

AI Tool Recommendation Matrix

Task	AI Tool	Human Oversight Level
Preliminary sizing	Aspen Plus Auto-Design	Low - verify results
Optimization (R, N, NF)	Genetic Algorithm	Medium - check constraints
Hydraulic design	ProMax/HYSYS	High - verify correlations
Cost estimation	ML-based CAPCOST	High - validate with quotes
Safety analysis	None - manual HAZOP	Critical - 100% human
Control system design	AI-assisted tuning	High - validate stability

Would you like me to develop a detailed optimization case study or create a process flow diagram showing the integration of this column with upstream/downstream units?

Prompt: Not at this time.

Results: You're welcome! Good luck with your vinyl chloride distillation column design project. If you need any further assistance with the design report or other aspects of the process engineering work, feel free to ask!

Khoa: ChatGPT 5 Thinking

Prompt: You are employed as a process engineering consultant at a chemical engineering design firm. Your team has been assigned to develop a preliminary design report for a new vinyl chloride monomer (VC) production facility to be built in Texas, USA. The planned plant will have a production capacity of 100,000 lb/hr of VC and will supply downstream polyvinyl

chloride (PVC) manufacturing operations. can you help me Draft process descriptions and PFD outlines • Suggest distillation column design parameters (stage count, reflux ratio, operating pressure) • Propose operating conditions or alternative column configurations • Suggest optimizations for energy integration or column sequencing? Make it short and simple and neat

Results: Scope & Assumptions (100,000 lb/h VCM)

- Technology: EDC → VCM thermal cracking with direct chlorination + oxychlorination loops to make EDC; full HCl recycle.
- Nominal rates (order-of-magnitude): ~140–160 klb/h EDC to cracking (tuned by yield/recovery), furnace outlet ~50–60% conversion per pass, VCM total recovery ≥98%.
- Inhibitor: trace O₂ injection; VCM handling at low T to suppress polymerization.

Process Description & PFD Outline

1. Chlorine + Ethylene (Direct Chlorination, DC) → EDC-DC (exothermic, liquid phase, catalyst).
2. Ethylene + HCl + O₂ (Oxychlorination, OX) → EDC-OX + H₂O (fluid bed copper catalyst, exothermic).
3. EDC Purification: Drying → Light-Ends Column (remove C₂/C₃, solvents) → Heavy-Ends Column (remove C₄+ tars) → EDC Storage.
4. EDC Cracking Furnace(s): 500–560 °C, ≈1–2 bar(g), short residence (10–30 s) → mix: VCM + HCl + uncracked EDC + traces.
5. Quench & HCl Recovery: Rapid quench → HCl/Org Separator → HCl to Absorber/Dryer → recycle to OX (or product).
6. VCM/EDC Separation: VCM Column (overhead VCM, bottoms EDC) → VCM to Dryer + Final Purification; EDC bottoms recycle to cracking.
7. VCM Finishing: Drying, inhibitor trim, cold condensation, product storage at low T.
8. Utilities & Treating: Chloride-bearing wastewater neutralization, off-gas treatment, vent recovery (refrigeration compressor), flare.

Suggested Distillation Column Design Parameters (starting points)

- Light-Ends EDC Column
 - Duty: remove light volatiles before furnace.
 - Stages: 20–30 (incl. condenser/reboiler)

- Reflux: 1.2–1.8
 - Pressure: 1.5–2.5 bar(a) (cooling-water condensation)
- Heavy-Ends EDC Column
 - Duty: protect furnace; cut high-bp/tars.
 - Stages: 25–35
 - Reflux: 1.5–2.2
 - Pressure: 1.2–1.8 bar(a)
- HCl/Org Decomposer or Stripper (if used post-quench)
 - Stages: 12–20
 - Pressure: 1.2–1.6 bar(a) (dry HCl overhead)
- VCM/EDC Column (Main Splitter)
 - Overhead: VCM (bp $-13.4\text{ }^{\circ}\text{C}$ at 1 atm); Bottoms: EDC (bp $83.5\text{ }^{\circ}\text{C}$).
 - Stages: 35–55 (VCM/EDC is a sharp, non-ideal split)
 - Reflux: 2.0–4.0 (tune vs. refrigeration)
 - Pressure options:
 - 1.3–1.8 bar(a): lower relative volatility loss, but needs refrigeration (-25 to $-5\text{ }^{\circ}\text{C}$).
 - 6–10 bar(a): chilled water or even cooling water condensation possible; check polymerization risk & compressor duty.
- VCM Finisher/Dehydrator (small polish column or dryer + guard bed)
 - Stages: 10–16 (if a polishing column is used)
 - Pressure: align with storage/transfer.

Operating Conditions (quick setpoints to start simulation)

- Direct Chlorination: $45\text{--}60\text{ }^{\circ}\text{C}$, 2–4 bar(a), FeCl_3 catalyst, slightly EDC-rich liquid phase.
- Oxychlorination: $200\text{--}250\text{ }^{\circ}\text{C}$ bed temp (adiabatic $220\text{--}260\text{ }^{\circ}\text{C}$), 2–4 bar(a), $\text{CuCl}_2/\text{Al}_2\text{O}_3$; control by quench boiler generating MP steam.
- Cracking Furnace: $500\text{--}560\text{ }^{\circ}\text{C}$ coil outlet; 0–2 bar(g); residence 10–30 s; quench within seconds.

- Quench: EDC quench to $<130\text{ }^{\circ}\text{C}$, then to $\sim 40\text{ }^{\circ}\text{C}$ prior to separations.
- VCM Product: -10 to $+5\text{ }^{\circ}\text{C}$ storage; inhibitor O_2 at ppm-level; N_2 pad.

Alternatives / Column Configurations

- VCM Column Heat Pump: Vapor recompression of overhead to drive reboiler (cuts refrigeration and steam).
- Split-Range Operation: Two VCM columns in parallel at different pressures for seasonal utility optimization.
- Side-Reboilers/Condensers: On VCM column to trim ΔT and refrigeration load.
- Dividing-Wall Column: Consider for EDC light/heavy ends if feed variability is high (capex vs. control complexity trade).
- Cracking Loop: Multi-furnace parallel trains with staggered decoking; consider small EDC flash upstream to remove dissolved light ends.

Energy Integration & Sequencing (quick wins)

- Recover furnace effluent sensible heat to preheat crude EDC or OX/DC feeds (approach $\Delta T \sim 10\text{--}20\text{ }^{\circ}\text{C}$).
- Use OX reactor waste heat boilers for MP steam; cascade to DC heater and column reboilers.
- Cold integration: subcool VCM product using returning refrigerant and cold overhead condensate.
- Consider mechanical vapor recompression on VCM overhead or absorption refrigeration driven by OX waste heat.
- Sequence columns so EDC purification precedes cracking (minimize fouling/coke), and VCM finisher last (spec insurance).

Safety & Operability Notes

- Keep VCM cold, inhibited, oxygen-monitored; eliminate dead-legs.
- Materials: chloride service (316L/Alloy 20 where needed), lined carbon steel elsewhere.
- Provide decoke provisions, quench reliability, HCl-resistant internals, and vent recovery to minimize flaring.

Comparison Table

	ASPEN	Copilot	Gemini	ChatGPT 5	Claude	ChatGPT 5 Thinking
Reflux ratio	0.11-0.17	2.5-3.5	1.2 x Rmin	2.0-3.0	1.5-3.0	2.0-4.0
Number of Stages	26-33	25-35	50-70	40-60	40-70	35-55
Feed Location	13-15	N/A	N/A	N/A	40-60% from bottom	N/A
Purity	98%-99%	N/A	>99.9%	Polymer-grade	>99.5%	N/A

Table 2: Summary of AI Design Parameters vs ASPEN

From this table, we can see that Claude was the only AI tool that was able to obtain the four indicated parameters. Despite that, the results from all AI bots seem to be very unoptimized, since the average number of stages suggested by these AI tools are higher than the results obtained from ASPEN while having about the same level of purity for the VC product.

Reflection

The Copilot AI had a straightforward approach in which the EDC is cracked and quenched and then separated from the VC. It also gave a range of reflux ratio and number of stages, but no feed location or VC purity in the product stream. Copilot AI also seemed to focus on the PFD outline, column specifications, and safety but not a lot of optimization and economics. Gemini AI had a more detailed process in which the operating pressure is higher than the rest and the number of stages is also the highest compared to the other AI responses. Gemini AI also had ideas about involving energy integrations into the process, such as pinch temp, as well as creating a code to render molecules. Unlike other models, ChatGPT 5 added a material balance as well as stating which EOS it's using for calculations. ChatGPT 5 also seemed to focus the most on energy integration, safety, and controls of the distillation systems. Claude approached the problem differently, where it focused more on the equipment sizing as well as optimizing the flowsheet. Lastly, ChatGPT 5 Thinking laid out a PFD outline with two operating pressure ranges for the process, and it also emphasized energy integration and safety of the process.

The creation of the PFD required us to have a critical understanding of the reaction thermodynamics and the process needed, with PFD being the blueprint to the process and the key to solving for the design correctly. It allows us to verbally communicate through a process, and understand the material and energy balances needed to be conducted. Based on the lecture material and our knowledge of process design, we carefully selected operation conditions that don't just work but also make sense.

For example, the realistic approach of the reactor operating at 1.5 atm and 90 C, while the blast furnace brought the temperature up to 500 C at 26 atm in order for complete vaporization. Moreover, the 1st distillation column had design conditions of 12 atm and 6 C, sequentially the 2nd distillation column had design conditions of 4.8 atm and 93 C. These values were chosen based on relevant information and industry conditions, to ensure optimum effectiveness in the separation and energy consumption of the process.

In simulation design via Aspen Plus, several assumptions were made for the simulation to converge. For the 1st column, the condenser pressure was set to 12 atm, and the reboiler pressure was set to 13 atm. This was to account for the loss of vapor and stability of the column. The HCl was the light key (LK) and VC was the heavy key (HK). This was done based on the relative boiling points/mass of the component. HCl had a 99.5% percent recovery in the distillate, and VC had a .5% recovery in the bottoms. The reflux ratio in Aspen Plus had an input of -1.15, representing the coefficient, which allows Aspen to know that the number of stages is unknown. The same was done for the 2nd column, but the condenser had an operating pressure of 4.8 atm, and the reboiler had a pressure of 5.2 atm. The LK was VC with 99.5% recovery, and the heavy EDC had a 0.5% recovery.

The use of AI tools in process design allowed engineers to quickly identify potential problems and resolve them. In this assignment, the AI was able to predict phase equilibria, recovery rates, and determine appropriate reflux ratios without strenuous trial and error in Aspen Plus. The AI models were able to create predictions on thermodynamics and the convergence behavior of a simulation; they were also able to differentiate logical and illogical operating conditions. AI also created flowsheets that are somewhat logical and relevant to our tasks. AI was able to complete challenging tasks that would take an individual engineer countless hours to identify and resolve. AI was able to do it in a matter of seconds, given the prompt, basis, and required details. AI was able to offer additional scope on the project by tabulating results in an Excel file or assisting in the development of an Aspen simulation. AI was not only taking into consideration the given task but also the greater and later tasks that would soon come.

AI has its downsides, such as its limited proficiency in certain areas of knowledge. Moreover, the use of AI in a complex process can cause inconsistencies in methodology. Since many plants rely on logic, but more importantly, experience that can only be gained by a human mind, using AI wouldn't be ideal in developing a plant. It's also been observed that AI tools often use a probabilistic method approach, meaning that since it can't solve everything at once, it will try to use probability to fill in information, which is non-ideal when developing an engineering process. Additionally, AI can't refer to the same values we would reference for the cost of utility of the plant, certain operating pressures, and operating conditions. In conclusion, the excess of AI reliance for optimization and control may deviate from standard process engineering decisions.

Bibliography

1. CHE 396 Lecture Notes
2. ASPEN Plus V14
3. AI Tools (ChatGPT, Gemini AI, Copilot, Perplexity, Claude)